Baidya Saha

Personal Webpage

L-272, Concordia University of Edmonton, 7128 Ada Blvd NW, Edmonton, AB T5B 4E4

+1 780 479 9310

baidya-nath-saha-55a40939

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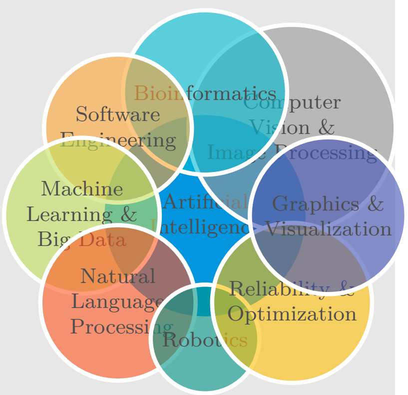
Baidya_Saha

BaidyaNathSaha

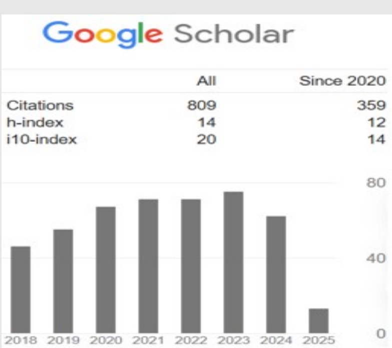
baidyasaha@gmail.com

Canadian

Research Interests



Research Citations



Research Interest Score: 528.7 Total Reads: 20,426

Personal Webpage

☐ http://baidyasaha.github.io

Current Affiliation

☐ Associate Professor, Information Technology, Concordia University of Edmonton (CUE)
☐ Chair (2021 - 2024), Mathematical and Physical Sciences, CUE, Canada

Education

2006-2011 **Ph.D. in Computer Science**
Department of Computing Science, University of Alberta, Edmonton, Canada
Thesis Title: The Evolution of Snake toward automation for multiple blob-object detection

2004-2006 **Master of Technology in Computer Science**
Indian Statistical Institute (I.S.I.), Kolkata, India
Thesis Title: A Regression based approach for leukocyte cell tracking

2002-2004 **Master of Technology in Quality, Reliability & Operations Research**
Indian Statistical Institute (I.S.I.), Kolkata, India
Thesis Title: Advanced statistical analysis for software effort estimation

1998-2002 **Bachelor of Mechanical Engineering**
Jadavpur University, Kolkata, India
Thesis Title: Simulation of a gear box design using 3D Studio Max

Employment

July, 19 - Pres **Associate Professor(July, 22 - Pres.), Assistant Professor(July, 19 - June, 22)**
Concordia University of Edmonton, Alberta, Canada
❖ Independent research, writing research grants and advising students
❖ Teaching and developing new courses for Bsc and Msc in IT Program
❖ Promoting AI and ML based industrial research through CIAR
❖ Was an MSc-IT program development committee member
❖ Contributed to a recruitment group to enhance diversity, boost domestic enrollment, and expand industry ties for the MISAM/MISSM program

July, 21 - Jun, 24 **Chair of the Mathematical and Physical Sciences, Program Director of MScIT program, and Coordinator of IT Program**
Concordia University of Edmonton, Alberta, Canada
❖ Graduate Studies Faculty Council Representative
❖ One of the representatives from Science in General Faculties Council (GFC)
❖ AI curriculum committee member, in partnership with NorQuest College
❖ Member of Faculty Hiring Committee (Math and Information Technology)
❖ Coordinator of Information Technology program
❖ Member of Science Faculty Council
❖ Co-Director (Academic) of Interdisciplinary Research Cluster on Machine Learning & Artificial Intelligence (IRCMLAI)
❖ One of the Concordia representatives for launching IT dual credit courses (IT 101 and IT 110) to Edmonton Catholic School District (ECSD)
❖ Expert reviewer of CUE's Program to Assist Grant Endeavours (PAGE)

Sep, 16 - Jun, 19 **Founding Member and Research Consultant**
Manufai SAS de CV, Escobedo, Mexico https://www.manufai.com
❖ Conducting industrial research towards product development
❖ Writing industrial research grants and patents
❖ Smart QC Sheet Digitalization: DL-based OMR/OCR system for QR/barcode detection and text recognition
❖ Industrial AR: Virtual welder training to enhance teaching and learning

Teaching Interests

Computer Vision

Machine Learning

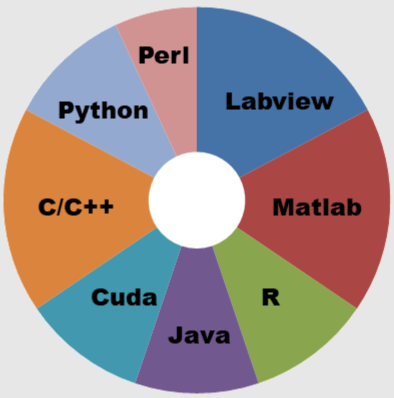
Big Data

Artificial Intelligence

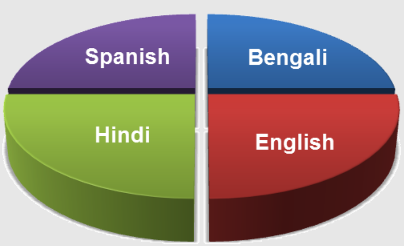
Parallel Computing

Numerical methods

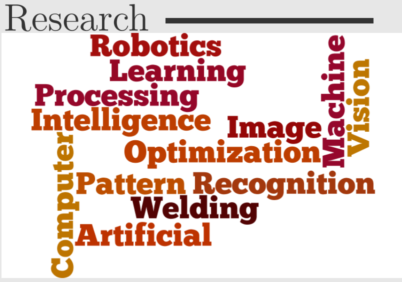
Programming



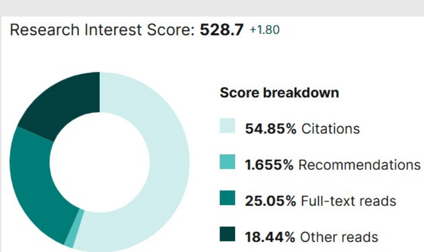
Languages



Research



ResearchGate Research Interest Score



Aug, 14 - Jun, 19 Assistant Professor

Centro de Investigación en Matemáticas (CIMAT), Mexico

❖ Independent research, advising students, writing research grants

❖ Communicating with industry, writing industrial research proposals and providing technical solutions to the industrial projects in the area of Image Processing, Statistics, Optimization, Machine Learning, and AI

❖ Teaching undergraduate and graduate courses

❖ Member of the committee that developed master program in Computational Statistics (selecting courses, syllabus and examination for the program). I developed syllabus for the courses of Programming, Parallel Computing, Machine Learning, and Artificial Intelligence

❖ Administrative and other responsibilities:

■ Master student recruitment committee member

■ Examining committee member for MS and PhD programs

Jul, 13 - Aug, 14 Postdoctoral Research Fellow

University of Calgary, Canada

❖ Classification of breast cancer tumor histology through morphometry

❖ Survival and recurrence analysis of stage I breast cancer with imaging biomarker and segregation of patients requiring chemotherapy

Nov, 11 - Jun, 13 Postdoctoral Research Associate

Wake Forest University School of Medicine, USA

❖ White matter lesion segmentation from brain MRI for Diabetes

❖ Three way (Alzheimer, Mild Cognitive Impairment and Cognitively Normal) classification for Alzheimer disease detection from brain MRI

❖ Prediction of brain age from pediatric brain MRI

Sep, 06 - Oct, 11 Teaching and Research Assistant

University of Alberta, Canada

❖ Developed novel techniques to automatically compute oil sand particle size distribution at various stages of oil sand fragmentation process that facilitate surface mining process

❖ Automatic detection of brain tumor and edema from brain MRI

Aug, 02 - Jul, 06 Research Assistant

Indian Statistical Institute

❖ Automatic detection and counting of rolling leukocytes from a vivo (mouse) cremaster muscle for the study of inflammation and the design of anti-/pro-inflammatory drugs

Proposals/Grants

Received

☐ NSERC Discovery and a Launch Supplement Grant (2020-2026): A unifying framework for integrating domain knowledge into machine learning algorithms for multidisciplinary industrial applications, $156,500.

☐ Mitacs Business Strategy Internship (BSI) Grant (2025): New User Enrollment System, $45,000.

☐ Concordia University of Edmonton Internal Seed Grant (2022): Secure privacy-preserving distributed machine learning algorithms for big data applications, $4,988.

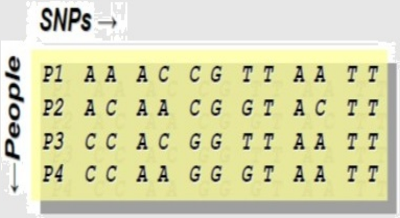
☐ Concordia University of Edmonton Internal Seed Grant (2021): Developing a novel Artificial Intelligence powered socially-aware recommender system for massive Online social network to filter the firehose of falsehood and propagation of misinformation, $4,400.

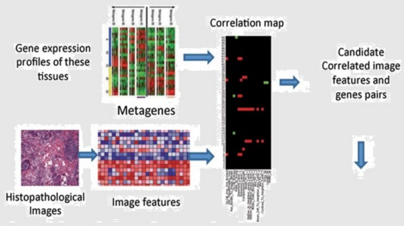
☐ Concordia University of Edmonton Special Call for Research Projects COVID-19 (SCRP COVID-19) (2020): Machine Learning based Rapid Self-Assessment and Understanding of COVID-19 Disease Progression to Support Public Policy Decisions, $2,350.

☐ NVIDIA Corporation Accelerated Data Science Call for Proposals (2019): Graphics Processing Unit (GPU) based acceleration for real time big data stream processing, two (2) Titan V GPU.

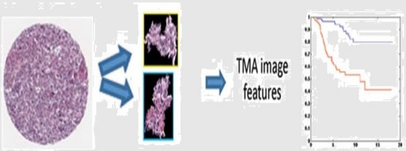
☐ Concordia University of Edmonton Internal Seed Grant (2019): Developing Efficient GPU based Differentiable End-to-End Deep Learning Architecture for Big Data Applications, $5,000.

Correlation between imaging and genomic biomarker



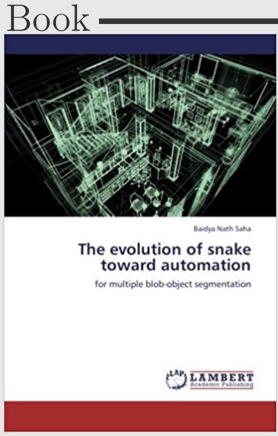


Training Phase: Discover the correlation between image and genomic biomarkers



Test Phase: Survival and recurrence Prediction with only relevant image features learned in the training phase

Book



Noted Publications

Conferences

❖ MICCAI
❖ ECML
❖ ACCV
❖ ICIP

Journals

❖ Pattern Recognition
❖ IEEE Signal Processing Letters
❖ Americal Journal of Neuroradiology
❖ Computerized Medical Imaging and Graphics

☐ Basic Science research fund, Conacyt, Mexico (2016-2019): Potential Association between image based phenotypes and genomic biomarkers for making therapeutic and prognostic decisions of cancer, $600,000.
☐ MITACS ACCELERATE internship with industrial partner AQL Management Consultants, Edmonton, Alberta, Canada (2010): Automatic passenger counting for Edmonton Light Rail Transit (LRT), Phase 1, $15,000.
☐ MITACS ACCELERATE internship with industrial partner AQL Management Consultants, Edmonton, Alberta, Canada (2010): Automatic passenger counting for Edmonton Light Rail Transit (LRT), Phase 2, $15,000.
☐ MITACS ACCELERATE internship with industrial partner AQL Management Consultants, Edmonton, Alberta, Canada (2009): Thermal image analysis for automatic non-invasive disease detection from Alberta Veterinary Surveillance Network, $15,000.

Award

❖ Prestigious NSERC discovery and a launch supplement grant as an early career researcher (2020-2026)
❖ Reduction in teaching award for research purposes by Concordia University of Edmonton (2020-2025)
❖ Member of the National System of Researchers (SNI), Mexico, level 1 (2015-2017)
❖ Basic science research grant as a young investigator (2016-2019)
❖ Post-doctoral fellowship offered from University of Calgary, Pittsburg, WFUSM, Mayo Clinic (2011-2014)
❖ MITACS industrial internship awards for three times (2009-2010)
❖ Several summer internship awards (2003-2011)
❖ Teaching and Research Fellowship (2002-2011)
❖ Mary Louise Imrie & J Gordin Kaplan Graduate Student Travel award, 2007

Teaching Experience

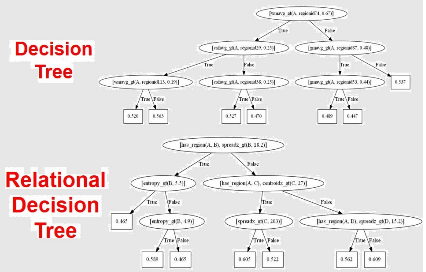
Curriculum Development

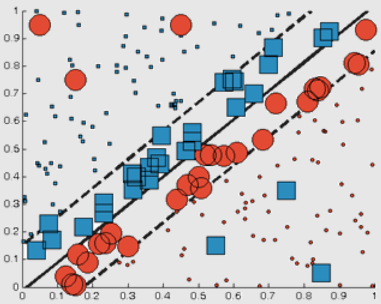
☐ Graduate / Undergraduate courses in Machine Learning and Artificial Intelligence: I developed syllabi for three courses, Machine Learning Tools and Techniques (IT 593), Machine Leaning Basics (IT 493), and Big Data (IT 596)at Concordia University of Edmonton (CUE).
☐ Master program in Computational Statistics was started at *Centro de Investigación en Matemáticas (CIMAT), Monterrey, Mexico* in August, 2015. I was actively involved in developing the curriculum for the master program. I prepared the syllabus for the courses: computational statistics, programming and analysis of algorithms, Big Data, and Artificial Intelligence.

Graduate Level

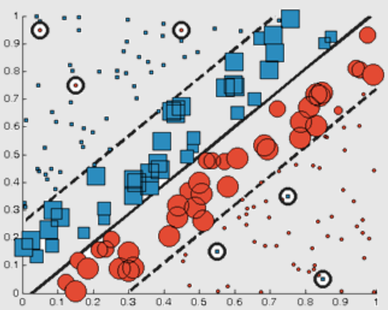
☐ Teaching at *Concordia University of Edmonton, Canada:*
❖ Fundamentals of Big Data: Tools and Algorithms (IT 596) in Winter 2022, 2023
❖ Machine Learning Tools and Techniques (IT 593) in Fall 2021, 2022, 2023, 2024
❖ IT Internship (IT 550) in Fall 2021, 2022
❖ IT Project (IT 581) in Spring 2021, 2022, 2023, 2024
❖ Modern Database Systems and Applications (IT 502) in Fall 2020, 2023
❖ Emerging Information Technologies (IT 572) in Fall 2020
❖ IT Project Management (IT 505) in Winter 2021, 2022, 2023
☐ Taught at *Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico:*
❖ Artificial Intelligence and Its Application in Manufacturing in Summer 2017
❖ Artificial Intelligence and Its Application in Robotics in Fall 2016
❖ Machine Learning and Big Data in Winter 2016
❖ Artificial Intelligence in Fall 2015
☐ Taught at *Centro de Investigación en Matemáticas (CIMAT), Monterrey, Mexico:*
❖ Fundamentals of Big Data: Tools and Algorithms in Fall 2018
❖ Numerical Methods and Optimization in Winter 2019, Winter 2018, Fall 2017
❖ Medical Image Processing in Winter 2018
❖ Computational Statistics in Fall 2017, 2018

Machine Learning





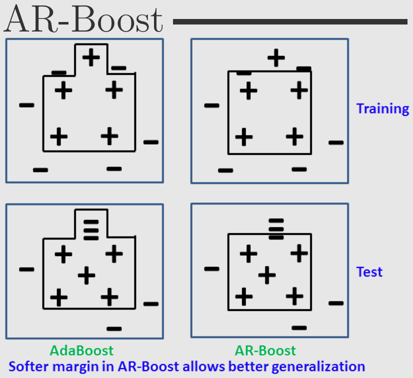
Normal SVM



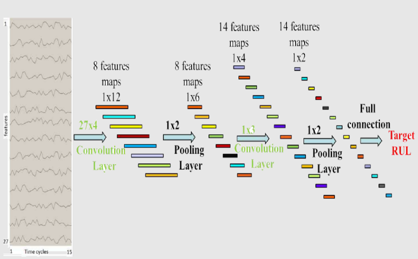
Proposed KNN-SVM

Increases margin by imposing less weight to the outliers and outliers are chosen by kNN

AR-Boost



Remaining Useful Life (RUL) Estimation



❖ Programming and Analysis of Algorithms in Fall 2016

Under graduate Level

☐ Teaching at *Concordia University of Edmonton, Canada:*

❖ Machine Leaning Basics (IT 493) in Winter 2022, Fall 2023, Fall 2024

❖ Senior Capstone Project (IT 451 & 452) in Fall 2020, 2024, Winter 2021, 2022, 2023, 2024

❖ Introduction to Software Development (CMPT 211) in Winter 2020

❖ Introduction to Information Technology (IT 101) in Winter 2020 and Fall 2019

❖ Introduction to Computing Science (CMPT 111) in Fall 2020, and 2019

☐ Worked as teaching assistant at *University of Alberta, Canada:*

❖ Numerical methods both in undergraduate (CMPUT 340/418) and graduate level (CMPUT 670) in Winter 2007, Fall 2008, and Fall 2009 respectively

❖ Introduction to Computing in Fall 2006

❖ 3D Graphics and Animation with 3DS Max in Winter 2011

❖ Optimizations in image analysis in Fall 2010

Training Received

☐ Attended Instructional Skill Workshop at *University of Calgary* in Fall 2013 :

❖ Acquired hands-on experience on designing mini-lessons using BOPPPS model

❖ Hands-on knowledge on the web-based e-learning technology called "Top-Hat"

❖ Knowledge of effective questioning for effective learning and creating a positive learning environment in the classroom

❖ Performed learning style survey before preparing lessons for large-room classes

☐ Attended in Project Management Workshop at *University of Alberta* in 2011 and learned :

❖ Strength Deployment Inventory (SDI) and card on the wall method

❖ Presentation skills by engaging audience through authenticity, and body language

❖ The art, science & practice of positive networking workshop conducted by MITACS

Research Contributions & Publications

My research interests lie at the intersection of developing and applying machine learning and computer vision algorithms. I actively contribute to a broad range of fields, including Artificial Intelligence, Machine Learning, Computer Vision, Robotics, Computational Cognitive Science, Reliability, Signal Processing, and Software Engineering.

To date, I have published 56 peer-reviewed research papers in reputed journals and conferences. My work has appeared in high-impact journals such as *Pattern Recognition* (Impact Factor: 7.5), *Engineering Applications of Artificial Intelligence* (IF: 7.5), *IEEE Signal Processing Letters* (IF: 3.2), and the *American Journal of Neuroradiology* (IF: 3.1). I have also presented at premier conferences including *MICCAI* (IF: 10.3, Acceptance Rate: ∼32%, ranked #1 in computer science), *ECML* (IF: 1.0, AR: ∼30%), *ACCV* (IF: 5.7, AR: ∼29%), and *ICIP* (IF: 7.5, AR: ∼45%).

My publications have received 802 citations on Google Scholar and over 19,129 reads on ResearchGate. These contributions span the following major research domains:

Books & Book Chapters

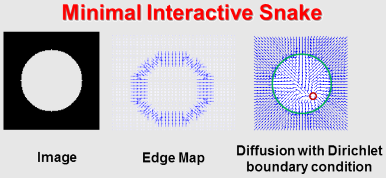
[1] B. N. Saha, *The evolution of snake toward automation for multiple blob-object segmentation.* LAP Lambert Academic Publishing, 2012, 1 google scholar citation.

[2] A. González-Cantú, M. E. Romero-Ibarguengoitia, and B. N. Saha, *Classification of Left Ventricular Hypertrophy and NAFLD Through Decision Tree Algorithm.* Chapman and Hall/CRC Press, Taylor & Francis Group, Editors Ayman El-Baz, Jasjit S. Suri, 2021.

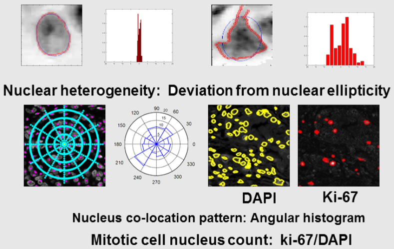
Advancement of Machine Learning Algorithms

One of my key research interests is integrating novel data-driven regularization strategies into classification algorithms such as AdaBoost, Support Vector Machines (SVM), and Decision Trees. These strategies aim to reduce test error by mitigating overfitting and promoting early convergence. In this context, I have developed two innovative algorithms: Adaptive

Snake

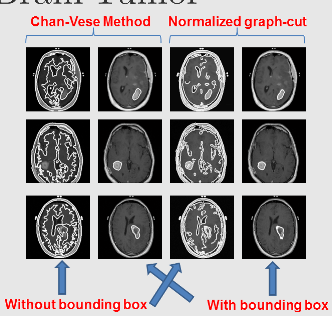


Biological Features

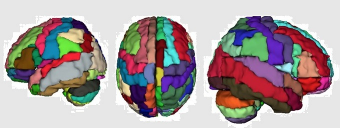


HierarchicalLevelSet

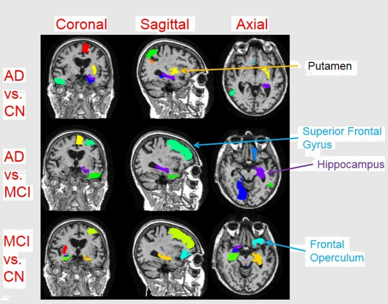
Brain Tumor



3D Brain Segemntation



Statistical Relational Learning (SRL)



Predictive segments automatically identified by our Statistical Relational Learning (SRL) algorithms for Alzheimer disease detection

Regularized Boosting (AR-Boost) and kNN regularized Support Vector Machine (kNN-SVM). Relevant publications are listed below.

[3] B. N. Saha, G. Kunapuli, N. Ray, J. A. Maldjian, and S. Natarajan, "Ar-boost: Reducing overfitting by a robust data-driven regularization strategy," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECMLPKDD)*. Springer, 2013, pp. 1–16, acceptance rate ~25%, 7 google scholar citations.

Computer Vision and Image Processing

My research focuses on advancing algorithms for segmentation, filtering, tracking, registration, object localization, detection, and validation—leveraging techniques such as Principal Component Analysis (PCA) and Adaptive Regularized Boosting (AR-Boost). I have developed several novel methods, including Probabilistic Quad Tree (PQT) and Fast Bounding Box (FBB) for approximate segmentation, automatic snake-based segmentation algorithms, Hierarchical Level Set (HLS), Topological Graph Cut (TGC), Robust Convergence Index (RCI), and the Inclusion Filter. Additionally, I have applied Statistical Relational Learning (SRL) for Alzheimer's disease prediction and designed a Minimum Spanning Tree (MST)-based sequence optimization algorithm for registering long, time-sequenced microscopic images. Relevant publications are listed below.

[4] A. Bahuguna, D. Yadav, A. Senapati, and B. N. Saha, "knn-svm with deep features for covid-19 pneumonia detection from chest x-ray," in *Mathematics and Computing*, B. Rushi Kumar, S. Ponnusamy, D. Giri, B. Thuraisingham, C. W. Clifton, and B. Carminati, Eds. Singapore: Springer Nature Singapore, 2022, pp. 103–115.

[5] T. A. Asif and B. N. Saha, "Instagram image filtration with computer vision," in *Proceedings of Intelligent Computing and Technologies Conference (ICTCon)*. AIJR, 2021, pp. 170–173.

[6] B. N. Saha, N. Ray, S. McArdle, and K. Ley, "Selecting the optimal sequence for deformable registration of microscopy image sequences using a two-stage mst-based clustering algorithm," in *Medical Image Computing and Computer Assisted Intervention - MICCAI 2017: 20th International Conference, Quebec City, QC, Canada, September 11-13, 2017, Proceedings*. Springer International Publishing, 2017, pp. 353–361, acceptance rate ~32%, 1 google scholar citation.

[7] ——, "A two-stage minimum spanning tree (mst) based clustering algorithm for 2d deformable registration of time sequenced images," in *Proceedings of IEEE International Conference on Image Processing (ICIP)*, 2017, pp. 1472–1476, acceptance rate ~40%, 1 google scholar citation.

[8] S. Natarajan, B. N. Saha, S. Joshi, A. Edwards, T. Khot, E. M. Davenport, K. Kersting, C. T. Whitlow, and J. A. Maldjian, "Relational learning helps in three-way classification of alzheimer patients from structural magnetic resonance images of the brain," *International Journal of Machine Learning and Cybernetics*, vol. 5, no. 5, pp. 659–669, 2014, impact factor 3.1, 19 google scholar citations.

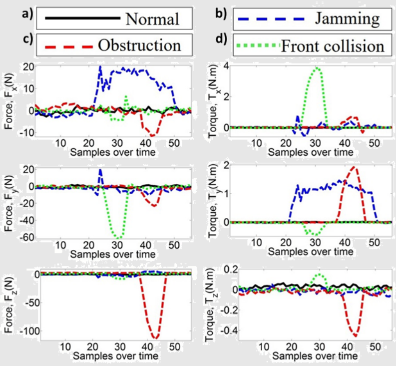
[9] J. A. Maldjian, C. T. Whitlow, B. N. Saha, G. Kota, C. Vandergriff, E. M. Davenport, J. Divers, B. I. Freedman, and D. W. Bowden, "Automated white matter total lesion volume segmentation in diabetes," *American Journal of Neuroradiology*, vol. 34, no. 12, pp. 2265–2270, 2013, impact factor 3.1, 49 google scholar citations.

[10] B. N. Saha, N. Ray, R. Greiner, A. Murtha, and H. Zhang, "Quick detection of brain tumors and edemas: A bounding box method using symmetry," *Computerized Medical Imaging and Graphics*, vol. 36, no. 2, pp. 95–107, 2012, impact factor 5.4, 162 google scholar citations.

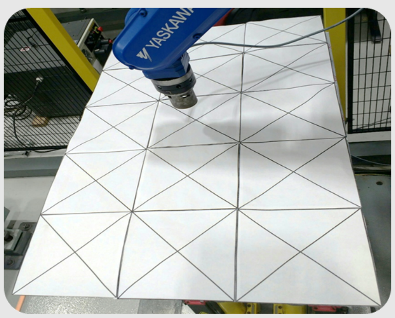
[11] B. N. Saha, N. Ray, and H. Zhang, "Snake validation: A pca-based outlier detection method," *IEEE Signal Processing Letters*, vol. 16, pp. 49–552, 2009, impact factor 3.2, 59 google scholar citations.

[12] B. N. Saha and N. Ray, "Image thresholding by variational minimax optimization," *Pattern Recognition*, vol. 42, pp. 843–856, 2009, impact factor 7.5, 91 google scholar citations.

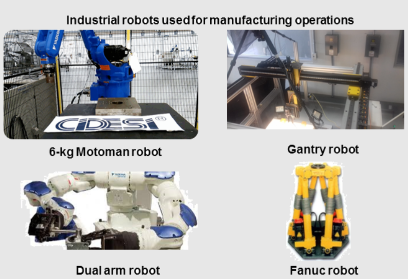
Robot Execution Failures for Peg-Hole Insertion Task

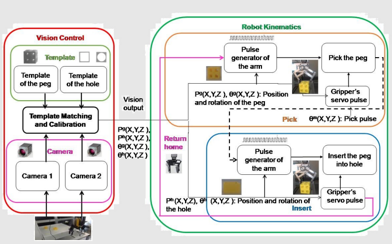


Signals of force torque sensors



Travelling salesman approach for hole search





Complete Block Diagram of Computer Vision Based Peg-Hole Insertion Task

[13] B. N. Saha, A. Saini, N. Ray, R. Greiner, J. Hugh, and M. Tambasco, "A robust convergence index filter for breast cancer cell segmentation," in *Proceedings of the IEEE International Conference on Image Processing (ICIP), 2014*. IEEE, 2014, pp. 922–926, acceptance rate ~40%, 1 google scholar citation.

[14] B. N. Saha, S. Natarajan, G. Kota, C. T. Whitlow, D. Bowden, J. Divers, B. I. Freedman, and J. A. Maldjian, "A novel hierarchical level set with ar-boost for white matter lesion segmentation in diabetes," in *Proceedings of the 11th International Conference on Machine Learning and Applications (ICMLA), 2012*, vol. 1. IEEE, 2012, pp. 90–95, 1 google scholar citation.

[15] S. Natarajan, S. Joshi, B. N. Saha, A. Edwards, T. Khot, E. Moody, K. Kersting, C. T. Whitlow, and J. A. Maldjian, "A machine learning pipeline for three-way classification of alzheimer patients from structural magnetic resonance images of the brain," in *Proceedings of the 11th International Conference on Machine Learning and Applications (ICMLA), 2012*, vol. 1. IEEE, 2012, pp. 203–208, 8 google scholar citations.

[16] S. Mukherjee, B. N. Saha, I. Jamal, R. Leclerc, and N. Ray, "A novel framework for automatic passenger counting," in *Proceedings of the 18th IEEE International Conference on Image Processing (ICIP)*, Brussels, Belgium, 2011, pp. 2969–2972, acceptance rate ~40%, 64 google scholar citations.

[17] B. N. Saha, N. Ray, and H. Zhang, "Automating snakes for multiple objects detection," in *Proceedings of the Asian Conference on Computer Vision (ACCV)*. Springer, 2010, pp. 39–51, acceptance rate ~29%, 6 google scholar citations.

[18] N. Ray, B. N. Saha, and H. Zhang, "Change detection and object segmentation: A histogram of features-based energy minimization approach," in *Proceedings of the IEEE Indian Conference of Computer Vision, Graphics and Image Processing (ICVGIP)*. IEEE, 2008, pp. 628–635, acceptance rate ~29%, 6 google scholar citations.

[19] N. Ray, B. N. Saha, and S. T. Acton, "Oil sand image segmentation using the inclusion filter," in *Proceedings of the 15th IEEE International Conference of Image Processing (ICIP)*. IEEE, 2008, pp. 2188–2191, acceptance rate ~45%, 3 google scholar citations.

[20] B. N. Saha, N. Ray, and H. Zhang, "Computing oil sand particle size distribution by snake-pca algorithm," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2008*. IEEE, 2008, pp. 977–980, acceptance rate ~48%, 10 google scholar citations.

[21] N. Ray and B. N. Saha, "Deformable object tracking: A kernel density estimation approach via level set function evolution," in *Proceedings of the International Conference on Pattern Recognition and Machine Intelligence (PReMI)*, 2007, pp. 624–631, acceptance rate ~25%.

[22] N. Ray, B. N. Saha, and M. Brown, "Locating brain tumor from mr imagery using symmetry," in *Proceedings of the Conference Record of the Forty-First Asilomar Conference on Signals, Systems and Computers, 2007. ACSSC 2007*. IEEE, 2007, pp. 224–228, acceptance rate ~40%, 52 google scholar citations.

[23] N. Ray and B. N. Saha, "Edge sensitive variational image thresholding," in *Proceedings of IEEE International Conference of Image Processing (ICIP)*, vol. 6. IEEE, 2007, pp. 37–40, acceptance rate ~45%, 52 google scholar citations.

Automation of Manufacturing Operations by Robots

To enhance industrial production efficiency, I developed computer vision-based solutions for autonomous parts mating, peg-in-hole insertion, and assembly operations. Our approach combines fast template matching with precise camera calibration to accurately identify workpieces (e.g., pegs and holes) in complex and occluded environments. We addressed position uncertainty by formulating it as a variant of the Traveling Salesman Problem and applied dynamic programming for optimal alignment. This system was successfully deployed on industrial robots, including the Motoman MH-6, SDA-20 dual-arm, and Cartesian gantry platforms, demonstrating robust performance in real-world manufacturing settings.

Additionally, I developed two signal classification algorithms—Bayesian Network (BN) and

Ontology based Valency Analyzer of Verbs for Anaphora Resolution

.... টেবিল থেকে [মিষ্টিটা] তোল, [ওটা] আমি খাব ....

.... take the [***sweet***] from the table, I will eat [***it***] ....

.... [টেবিল] থেকে মিষ্টিটা তোল, [ওটা] আমি ধোব ....

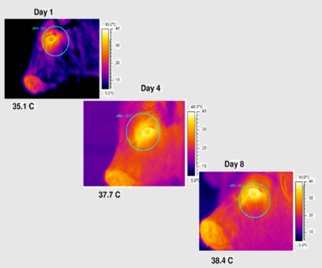
.... take the sweet from the [***table***], I will wash [***it***] ....

Evaluation Metrics for Anaphora Resolution System

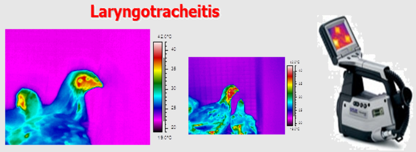
Metric		Baseline	System I	System II
MUC	P	0.437	0.489	0.538
	R	0.426	0.462	0.605
	F1	0.431	0.475	0.569
B^3	P	0.577	0.678	0.740
	R	0.676	0.832	0.842
	F1	0.608	0.747	0.787
CEAFM	P	0.614	0.654	0.786
	R	0.602	0.646	0.695
	F1	0.608	0.650	0.737
CEAFE	P	0.661	0.797	0.804
	R	0.601	0.642	0.555
	F1	0.630	0.712	0.656
BLANC	P	0.480	0.542	0.765
	R	0.582	0.678	0.771
	F1	0.526	0.603	0.767
Avg.	F1	0.560	0.637	0.703

Thermography

Bovine viral diarrhea



Laryngotracheitis

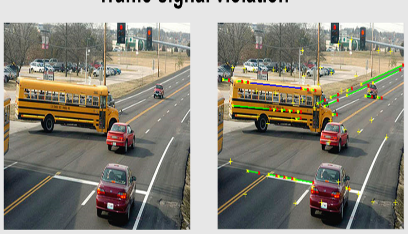


Non-invasive disease detection

People Detection

Intelligent Transportation

Traffic signal violation



Autoregressive Dynamic Learning - to classify force and torque signals corresponding to four failure types: insertion failure, obstruction, jamming, and frontal collision. To further optimize the insertion process, we implemented Gaussian Mixture Regression (GMR)-based impedance control. For solving the robot inverse kinematics problem, I enhanced the Genetic Algorithm (GA) by embedding domain knowledge into all key stages—initialization, selection, crossover, and mutation—achieving faster and more reliable convergence.

These methodologies have been validated through successful deployment on industrial robotic systems. The associated research has been published in peer-reviewed venues, as listed below.

[24] D. Ortega-Aranda, J. F. Jimenez-Vielma, B. N. Saha, and I. Lopez-Juarez, "Dual-arm peg-in-hole assembly using dnn with double force/torque sensor," *Applied Sciences*, vol. 11, no. 15, 2021, 13 google scholar citations. [Online]. Available: https://www.mdpi.com/2076-3417/11/15/6970

[25] D. Ortega-Aranda, I. Lopez-Juarez, B. N. Saha, R. Osorio-Comparan, M. P. na Cabrera, and G. Lefrnc, "Learning contact states during peg-in-hole assembly with a dual-arm robot," in *Proceedings of the IEEE CHILECON*, 2017, pp. 1–6, 17 google scholar citations.

[26] A. S. Cienfuegos, B. N. Saha, J. Romero-Hdz, and D. Ortega, "Efficient integration of template matching , calibration and triangulation for automating peg hole insertion task using two cameras," *SSRG International Journal of Computer Science and Engineering (SSRG-IJCSE)*, vol. 3, pp. 61–70, Nov. 2016, impact factor 2.150, 1 google scholar citation.

[27] E. Rodriguez, B. N. Saha, J. Romero-Hdz, and D. Ortega, "A multi-objective differential evolution algorithm for robot inverse kinematics," *SSRG International Journal of Computer Science and Engineering (SSRG-IJCSE)*, vol. 3, pp. 71–79, Nov. 2016, impact factor 2.150, 13 google scholar citations.

[28] J. Alonso-Tovar, B. N. Saha, J. Romero-Hdz, and D. Ortega, "Bayesian network classifier with efficient statistical time-series features for the classification of robot execution failures," *SSRG International Journal of Computer Science and Engineering (SSRG-IJCSE)*, vol. 3, pp. 80–89, Nov. 2016, impact factor 2.150, 4 google scholar citations.

[29] A. Sauceda, E. Rodriguez, J. Romero, D. Ortega, and B. N. Saha, "Implementation of computer vision guided peg-hole insertion task performed by robot through labview," *MICAI 2016, Part I, Lecture Notes in Artificial Intelligence (LNAI) 10061*, pp. 437–458, 2017, 1 google scholar citation.

Automation and Optimization of Welding Process

Welding deformation presents significant challenges in metal joining, often leading to design constraints, increased rework, elevated quality control costs, and higher capital expenditures. Optimizing the welding sequence is critical to minimizing deformation and residual stress; however, this task is computationally intensive due to its complex combinatorial nature.

To tackle this, I implemented a suite of optimization techniques—including a Modified Lowest Cost Search (MLCS), a multi-objective Genetic Algorithm (GA), and a Reinforcement Learning (RL) framework—to generate pseudo-optimal welding sequences that effectively reduce deformation and residual stress. Notably, our research was the first to demonstrate the combined use of Low Temperature Transform (LTT) and conventional wire for welding sequence optimization, resulting in a significant improvement in weldment fatigue life.

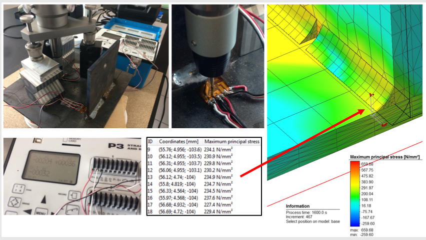
These contributions have been validated through experimental studies and documented in the peer-reviewed publications listed below.

[30] J. Romero-Hdz, B. N. Saha, N. Hofbauer, and P. del C. Z. Robledo, "Os-weldsim: An open-source framework for finite element-based welding simulation," *The International Journal of Advanced Manufacturing Technology (Submitted)*, 2025.

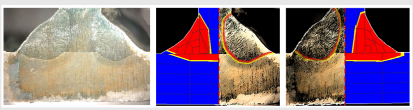
[31] J. Romero-Hdz, B. N. Saha, S. Tstutsumia, R. Fincatoa, and G. Toledo, "Incorporating domain knowledge into reinforcement learning to expedite welding sequence optimization," in *Engineering Applications of Artificial Intelligence*, vol. 91, 2020, pp. 1–10, 14 google scholar citations.

Welding Simulation

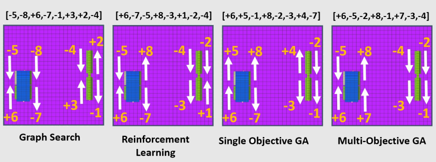
Welding stress validation through hole drilling method



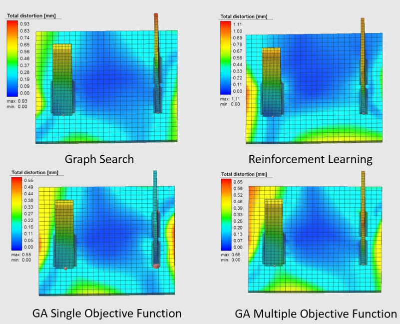
Residual Stress comparison between real (hole drilling method) and simulation experiment conducted through Simufact®software



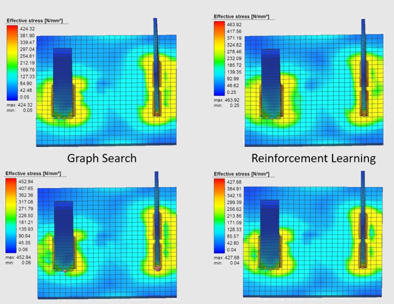
the heat source model of the welding simulation was validated through macroetch test



Welding sequence optimization



Structural deformation due to welding



Residual stress due to welding

[32] ——, "Statistical validation of weldment deformation through welding simulation and 3d optical scanning," in *Proceedings of the Welding Structure Symposium*, 2017, pp. 1–8.

[33] J. Romero-Hdz, S. Tstutsumia, R. Fincatoa, and B. N. Saha, "Influence of welding sequence on residual stress and deformation pattern using conventional and ltt wires," in *Proceedings of the Welding Structure Symposium*, 2017, pp. 1–8.

[34] J. Romero-Hdz, B. N. Saha, G. Toledo, and I. Lopez, "A reinforcement learning based approach for welding sequence optimization," *Transactions on Intelligent Welding Manufacturing*, pp. 33–45, 2017, 13 google scholar citations.

[35] J. Romero-Hdz, B. N. Saha, and G. Toledo, "Welding sequence optimization through a modified lowest cost search algorithm," *Computer Science and Engineering*, vol. 6, pp. 25–32, 2016, 5 google scholar citations.

[36] J. Romero-Hdz, S. Aranda, G. Toledo-Ramirez, J. Segura, and B. N. Saha, "Deformation and residual stress based multiobjective genetic algorithm for welding sequence optimization," *Journal of Research in Computing Science*, vol. 132, pp. 155–179, 2017.

[37] J. Romero-Hdz, S. Aranda, G. Toledo, J. Segura, and B. N. Saha, "An elitism based genetic algorithm for welding sequence optimization to reduce deformation," *Journal of Research in Computing Science*, vol. 121, pp. 17–36, 2016, 18 google scholar citations.

[38] J. Romero-Hdz, B. N. Saha, G. Toledo-Ramirez, and D. Beltran-Bqz, "Welding sequence optimization using artificial intelligence techniques, an overview," *SSRG International Journal of Computer Science and Engineering (SSRG-IJCSE)*, vol. 3, pp. 90–95, Nov. 2017, impact factor 2.150, 19 google scholar citations.

[39] J. Romero-Hdz, B. N. Saha, and G. Toledo, "Deformation driven fast and approximate shortest path algorithm for selecting a pseudo-optimal welding sequence," *Global Conference on Engineering and Applied Science*, vol. GCEAS-487, pp. 439–449, 2016.

[40] J. Romero-Hdz, G. Toledo, and B. N. Saha, "Deformation and residual stress based multiobjective genetic algorithm for welding sequence optimization," in *2016 Fifteenth Mexican International Conference on Artificial Intelligence (MICAI)*, Oct 2016, pp. 80–91, 11 google scholar citations.

Reliability and Optimization for Predictive Maintenance

Prognostics have gained increasing attention due to the critical need for accurate estimation of Remaining Useful Life (RUL) in diverse applications. My research focuses on developing machine learning-based prognostic models for both electric motors and cancer patients, with the goal of improving predictive accuracy and reliability.

In parallel, I have worked on optimizing design parameters for life testing plans under censoring schemes using genetic algorithm-based multi-objective optimization. A particular focus has been placed on progressive censoring, a complex combinatorial optimization problem. To address its computational challenges, I introduced a novel restricted weak integer composition-based sample initialization algorithm, which significantly accelerates convergence and enhances the robustness of the Genetic Algorithm.

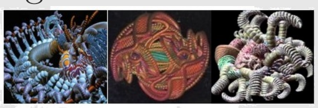
These advancements contribute to more efficient and reliable prognostic modeling and testing strategies. Related publications are listed below.

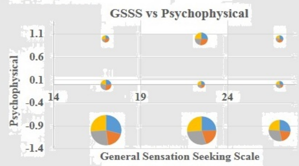
[41] R. Bhattacharya, B. N. Saha, G. González-Farías, and N. Balakrishnan, "Multi-criteria based optimum life testing plans under hybrid censoring," *Test*, pp. 1–24, 2019, 14 google scholar citations.

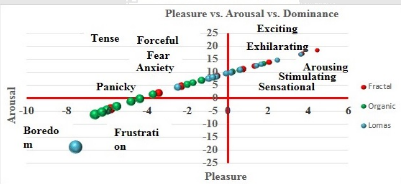
Computational Cognitive Science

One of my research areas focuses on developing mathematical models and computational simulations to deepen our understanding of human perception, cognition, learning, and behavioral psychology. As part of this work, I have designed novel statistical models and introduced a new statistical visualization technique - the pie-bubble chart - to analyze complex perceptual data.

Cognitive Science







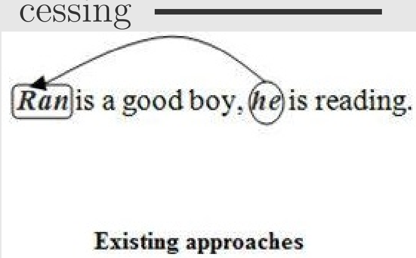
My research addresses key topics such as color perception, the influence of digital diversion in mitigating curriculum-related stress, and the perceptual characteristics of computer-generated biological imagery that trigger horripilation (goosebumps). These interdisciplinary efforts bridge computational modeling with psychological theory, offering new insights into human responses to digital stimuli.

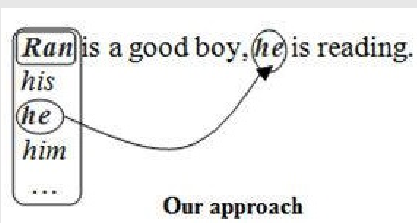
The findings from this work are detailed in the publications listed below.

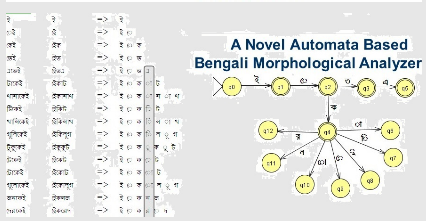
[42] T. P. Mukhopadhyay, B. N. Saha, N. Gurieva, and R. T. Lopez, "Rosa mexicano: the social optics of a color neologism," *Journal of the International Colour Association*, pp. 1–16, 2017.

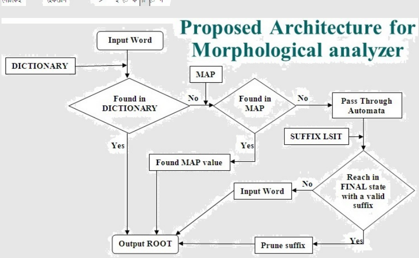
[43] T. P. Mukhopadhyay, M. Siprut, and B. N. Saha, "Digital diversions in education: Interactive multimedia for adolescent motivation in unilateral classroom scenarios," *International Journal of Innovation, Creativity and Change*, vol. 3, pp. 60–74, 2017.

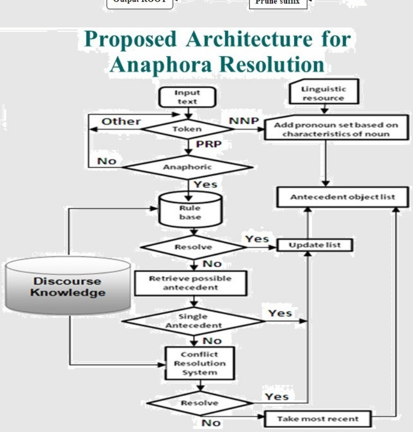
Natural Language Processing











Natural Language Processing in Resource-Scarce Languages

My research in Natural Language Processing (NLP) focuses on developing innovative solutions for key challenges such as anaphora resolution, sentiment analysis, fake news detection, hate speech and offensive content identification, and text summarization, particularly for resource-scarce languages. These languages present unique challenges due to limited annotated datasets and significant linguistic diversity.

To address these challenges, we employ strategies such as transfer learning, multilingual models, and community-driven data collection. For example, our work on anaphora resolution in Bengali integrates linguistic domain knowledge to enhance traditional machine learning models like Support Vector Machines, Decision Trees, and AdaBoost. We have also developed deep learning frameworks—including LSTM-based architectures—for sentiment analysis and hate speech detection in low-resource Indo-European languages using social media data.

These efforts are aimed at improving the generalization capabilities of NLP models across diverse linguistic contexts and contributing to the preservation and digital representation of underrepresented languages. Our research bridges computational methods with linguistic insights to support the development of inclusive and effective NLP tools.

Publications in this area are listed below.

[44] A. Bahuguna, D. Yadav, A. Senapati, and B. N. Saha, "A unified deep neuro-fuzzy approach for covid-19 twitter sentiment classification," *Journal of Intelligent & Fuzzy Systems*, vol. 42, no. 5, pp. 4587–4597, 2022, 5 google scholar citations.

[45] B. N. Saha, A. Senapati, and U. Garain, "A deep learning framework for anaphora resolution from social media text," in *Advanced Machine Intelligence and Signal Processing*, D. Gupta, K. Sambyo, M. Prasad, and S. Agarwal, Eds. Singapore: Springer Nature Singapore, 2022, pp. 687–695, 1 google scholar citation.

[46] A. A. Ayandeyi and B. N. Saha, "Twitter data sentiment analysis to understand effect of covid-19 on mental health," in *Proceedings of Intelligent Computing and Technologies Conference (ICTCon)*. AIJR, 2021, pp. 174–179, Best session paper award.

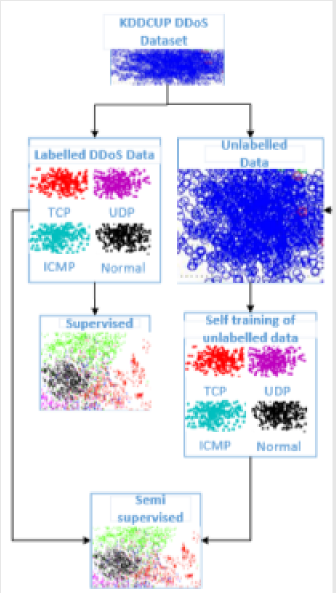
[47] B. N. Saha and A. Senapati, "Hate speech and offensive content identification: Lstm based deep learning approach @ hasoc 2020," in *Working Notes of FIRE 2020 - Forum for Information Retrieval Evaluation*, vol. 2826, 2020, pp. 290–297.

[48] ——, "Long short term memory (lstm) based deep learning for sentiment analysis of english and spanish data," in *International Conference on Computational Performance Evaluation (ComPE)*, 2020, pp. 1–5, 34 google scholar citations.

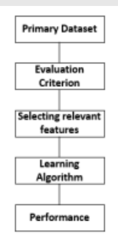
[49] B. N. Saha, A. Senapati, and A. Mahajan, "Lstm based deep rnn architecture for election sentiment analysis from bengali newspaper," in *International Conference on Computational Performance Evaluation (ComPE)*, 2020, pp. 1–6, 11 google scholar citations.

[50] A. Senapati, A. Poudyal, P. Adhikary, S. Kaushar, A. Mahajan, and B. N. Saha, "A machine learning approach to anaphora resolution in nepali language," in *International*

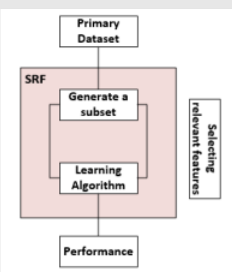
Network Security



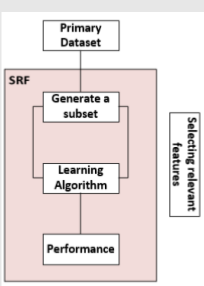
ML and Network Security



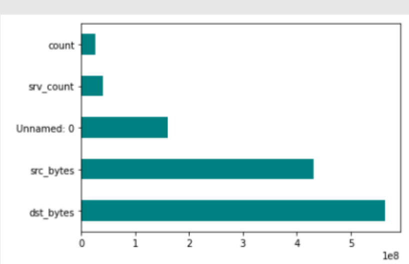
Filter Method



Wrapper Method



Embedded Method



Feature Selection through SelectKBest

Conference on Computational Performance Evaluation (ComPE), 2020, pp. 1–6, 5 google scholar citations.

[51] B. N. Saha and A. Senapati, "Lstm based deep rnn architecture for hate speech and offensive content (hasoc) identification in indo–european languages," in *The 11th meeting of Forum for Information Retrieval Evaluation (FIRE)*, 2019, 6 google scholar citations.

Others:

Signal Processing

This research proposes a deep learning-based end-to-end architecture for speech recognition, designed to improve both accuracy and efficiency across diverse linguistic contexts. The architecture integrates multiple layers of neural networks to directly process raw audio input, eliminating the need for manual feature extraction and simplifying the recognition pipeline. By leveraging advanced deep learning models, the system effectively learns complex speech patterns, enhancing performance in noisy environments and enabling robust recognition across various speech types.

[52] R. A. de Morais and B. N. Saha, "End-to-end speech recognition using recurrent neural network (rnn)," in *Proceedings of Intelligent Computing and Technologies Conference (ICTCon)*. AIJR, 2021, pp. 152–158, acceptance rate ~40%, 1 google scholar citation, Second best session paper award.

Network Security

Distributed Denial-of-Service (DDoS) attacks pose significant threats to financial institutions by overwhelming networks and disrupting services. To counter these threats, machine learning (ML) offers predictive capabilities to identify and mitigate potential DDoS attacks. In the study *"Using Machine Learning to Predict Distributed Denial-of-Service (DDoS) Attack"* by Qozeem Adeniyi Adeshina and Baidya Nath Saha, published in the *Proceedings of the Intelligent Computing and Technologies Conference (ICTCon2021)*, the authors explore the development of ML classifiers to detect DDoS attacks before they occur.

Using the KDD-99 dataset, a comprehensive network traffic dataset, the researchers trained various ML classifiers to distinguish between normal and DDoS traffic across protocols like ICMP, TCP, and UDP. The study employed seven distinct feature selection techniques and applied them to ten different ML classifiers to enhance attack detection accuracy. The findings demonstrated that certain classifiers could predict DDoS attacks effectively within short timeframes, providing valuable insights for preemptive security measures.

Integrating these machine learning models into the cybersecurity infrastructure of financial institutions can significantly enhance defenses against DDoS attacks. By accurately predicting and identifying malicious traffic patterns, these models enable timely responses, ensuring service continuity and the protection of financial assets. This research highlights the potential of machine learning in strengthening cybersecurity measures within the financial sector.

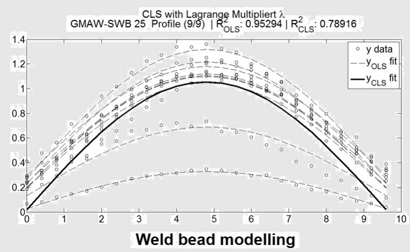
[53] M. D. M. Islam, C. S. P. B. N. Saha, S. Parvin, W. M. Abdullah, and K. F. Hasan, "A privacy-preserving trustworthy behavioral authentication system," *ACM Transactions on Privacy and Security (TOPS) (Submitted)*, 2025.

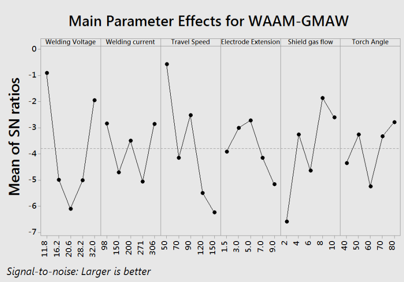
[54] Q. A. Adeshina and B. N. Saha, "Using machine learning to predict distributed denial-of-service (ddos) attack," in *Proceedings of Intelligent Computing and Technologies Conference (ICTCon)*. AIJR, 2021, pp. 159–169.

Internet Of Things

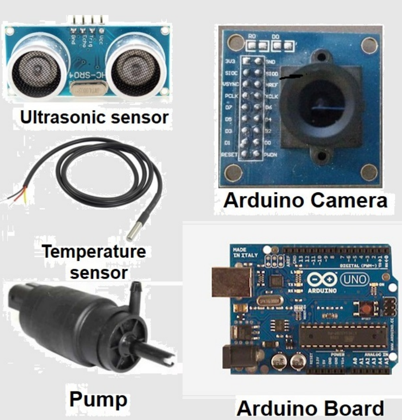
This research introduces a machine learning-driven Internet of Things (IoT) smart healthcare kit designed to enhance patient care through continuous online monitoring of vital health indicators. The study also explores diagnostic applications, utilizing K-Nearest Neighbor (K-NN) and Random Forest classifiers for heart disease prediction based on pathology data. Additionally, the VGG-19 deep learning architecture is employed for lung disease detection from X-ray images. These machine learning models support physicians in accurate disease diagnosis, ultimately improving patient outcomes.

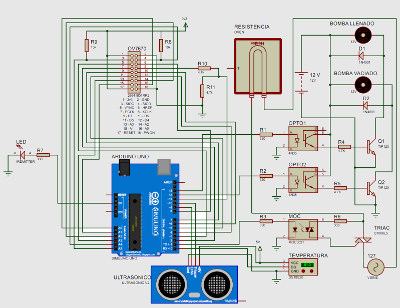
Wire + Arc Additive Manufacturing (WAAM)





Water pump controller





This research highlights how integrating machine learning with IoT technologies can revolutionize healthcare delivery by enabling remote monitoring and early disease detection.

[55] L. Narayanagari and B. N. Saha, "Machine learning driven iot based smart health care kit," in *Proceedings of Intelligent Computing and Technologies Conference (ICTCon)*. AIJR, 2021, pp. 180–185, acceptance rate ~40%, 2 google scholar citations.

Software Engineering

This research proposes a machine learning-based approach for software test case selection, designed to optimize regression testing with limited resources. By leveraging machine learning models, the study improves the efficiency of test case selection, enabling software testing teams to prioritize the most critical cases. The approach involves data preprocessing, including converting categorical data into numerical values, and applyingnatural language processing (NLP) techniques to extract relevant features from test case titles.

Experimental results show that this machine learning-driven method significantly reduces manual efforts by automating test case selection, leading to substantial time and resource savings while enhancing overall testing efficiency.

[56] V. Cheruiyot and B. N. Saha, "A machine learning based approach for software test case selection," in *Proceedings of Intelligent Computing and Technologies Conference (ICTCon)*. AIJR, 2021, pp. 186–189, acceptance rate ~40%, 1 google scholar citation.

Long Term Research Goals

In the long term, my research interests are centered on solving application-driven, large-scale big data analysis challenges. I believe that while engineering solutions are essential, they are insufficient on their own to address such complex problems in real-world applications. Theoretical advancements are crucial for developing optimal solutions to these challenges.

Citations of my publications are available via the following bibliometrics link: Google Scholar Citations

Industry Experience

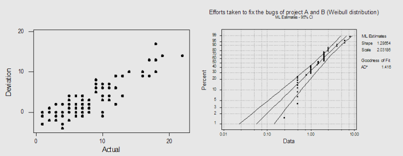
2014 - 2019 Research Consultant
CIMAT, Monterrey, Mexico

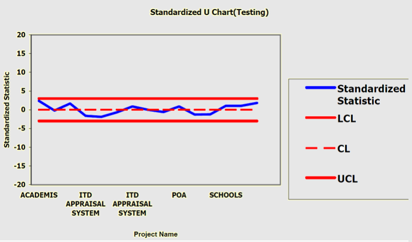
- ☐ *Automated quality inspection of the painted surface of the rearview mirrors of the automobiles through laser ray and camera*
- ☐ *Monitoring the performance and controlling the temperature of the industrial furnace with minimum reosurces and cost through thermodynamic modelling and evolutionary Markov Chain Monte Carlo (MCMC)*
- ☐ *Hidden Markov Model for the temperature control inside a specialized furnace to improve the production quality of the core of the transformers*
- ☐ *Monte Carlo simulations for probabilistic seismic loss assessment*
- ☐ *A Bayesian time series control chart and two stage sampling technique for visual inspection of the electronic components*
- ☐ *A response surface methodology and Taguchi orthogonal design for improving the quality of cold metal welding for making industrial steel tubes*

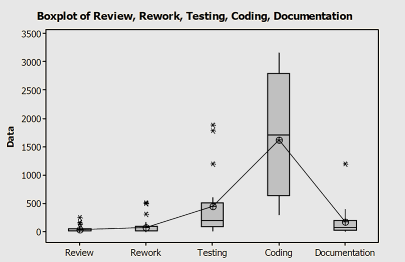
2013-2014 Research Consultant
Acellere, Pune, India

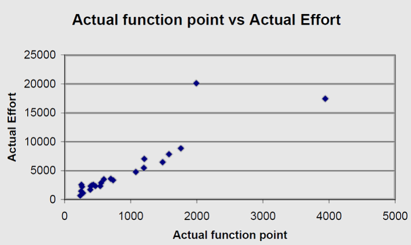
- ☐ *Assessing the quality of the class of software on the basis of a set of metrics and developing a new metric by incorporating the class dependency information*

Software Effort Estimation









2009-2010	Summer Intern	

2009-2010 Summer Intern
AQL Management Consulting Inc., Edmonton, Alberta, Canada
☐ Conducted several image processing research projects including
❖ Developed an autonomous passenger counting system for LRT (Light Railway Transit)
❖ Built an automated tool to analyze chickens behavior monitoring
❖ Early disease detection of flocks using infrared imaging
❖ Automatic assessment of feather cover of laying hens by infrared thermography

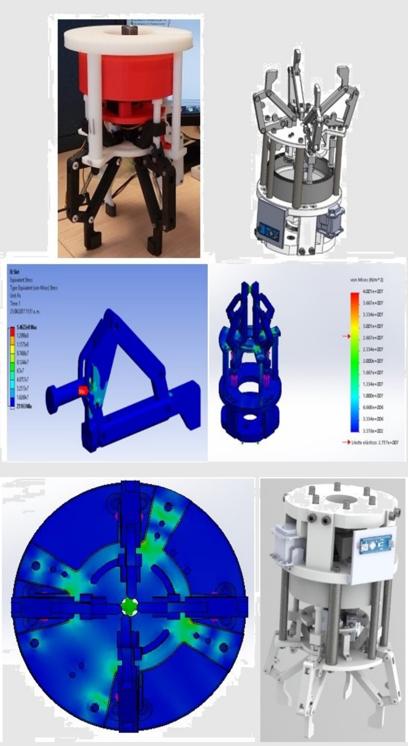
2005 Summer Intern
Cognizant Technology Solutions, Kolkata, India
☐ Developed a game theory based Economic Modelling of Optimal Bidding in one client, multiple vendors' scenario in a win-win condition

2004 Summer Intern
Usha Martin Limited, Kolkata, India
☐ Developed Software tool to facilitate Supply Chain Management System

2004 Summer Intern
ITC InfoTech India Limited , Bangalore, India
☐ Conducted several statistical software quality projects including
❖ A Bayesian approach to evaluate methodology followed to establish baselines of build efforts on the basis of SAC (Simple/Average/Complex) methodology
❖ Developed software defect management system

2003 Summer Intern
Robert Bosch India Limited, Bangalore, India
☐ Conducted several statistical data analysis & operations research projects including
❖Estimated the efforts required to fix the number of bugs reported by the customer
❖ Developed methodology for managing project schedule whenever it overruns
❖ Measured statistical Customer satisfaction index (CSI) based on Customer Survey and Delivery delay

Gripper design using FEM



Collaborators

❖ Prof. Nilanjan Ray, University of Alberta, Canada
❖ Prof. Joseph Maldjian, University of Texas Southwestern Medical Center
❖ Prof. Klaus Ley, La Jolla Institute for Allergy & Immunology, USA
❖ Prof. Mauro Tambasco, San Diego State University
❖ Prof. Ross Mitchell, University of Alberta, Canada
❖ Prof. Sriraam Natarajan, University of Texas at Dallas, USA
❖ Prof. Christopher Whitlow, Wake Forest School of Medicine, USA
❖ Prof. Russell Greiner, University of Alberta, Canada
❖ Prof. Tirtha Prasad Mukhopadhyay, University of Guanajuato, Mexico
❖ Prof. Dipti Prasad Mukhopadhyay, Indian Statistical Institute, Kolkata, India
❖ Prof. Md Morshedul Islam, Concordia University of Edmonton, Canada
❖ Prof. Wali Mohammad Abdullah, Concordia University of Edmonton, Canada

Students & Thesis Supervision

To date, I had the opportunity to mentor seventy (70) trainees - comprising 2 postdoctoral fellows, 2 PhD students, 39 master's (thesis-based) students, and 19 undergraduate research assistants—from diverse disciplines and institutions. This extensive mentorship has resulted in 54 peer-reviewed publications, many co-authored with trainees. Beyond research supervision, I have actively supported trainees' professional development, including mentoring in communication, CV writing, and career preparation. Notable outcomes include academic appointments for both postdoctoral fellows, startup co-founding by a PhD trainee, and several prestigious national and provincial awards for graduate and undergraduate students (e.g., NSERC CGS-M, USRA, Alberta Innovates GSS, and Governor General's Academic Gold Medals). My consistent commitment to excellence in training has cultivated a pipeline of highly qualified personnel poised for impactful careers in academia, industry, and innovation.

Postdoctoral Researcher

Sep, 16 - Jan, 19 Ritwik Bhattacharya

PhD, Indian Statistical Institute

☐ Project title: *Optimization problems in the domain of Reliability*

❖ Present Position: *Assistant Professor, The University of Texas at El Paso*

May, 18 - May, 19 Apurbalal Senapati

PhD, Indian Statistical Institute

☐ Project title: *Developing deep learning algorithms for Anaphora Resolution and Sentiment Analysis in Natural language Processing (NLP)*

❖ Present Position: *Assistant Professor, Central Institute of Technology, Kokrajhar, Assam, India*

Doctoral Students

Sep, 15 - Dec, 21 Jesus Antonio Romero Hernandez

Doctorate program in Mechatronics

Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico

☐ Thesis title: *Development of methodologies for reducing distortions in industrial welding processes through optimization with FEM analysis and artificial intelligence techniques*

❖ Present Position: *CEO, Manufai, Monterrey, México*

Jan, 16 - Dec, 21 M. C. David Ortega Aranda

Doctorate program in Mechatronics

Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico

☐ Thesis title: *Study and experimentation of artificial intelligence algorithms to learn bimanual tasks in a double arm robot*

❖ Present Position: *Technology Innovation Engineer - Electronics, Baker Hughes*

Master Students

Sep, 24 - Dec, 25 Daxit Himmatbhai Surani, Recipient of Alberta Innovates Graduate Student Scholarship (GSS), $17,000

Master of Science (MSc) in Information Technology (In Progress)

Concordia University of Edmonton (CUE), Alberta, Canada

☐ Thesis title: *SMART-MED: Scalable Multimodal Reinforcement-Driven AI for Medical Diagnostics in Alberta*

❖ Present Position: *MSc Student, CUE*

Sep, 24 - Dec, 25 Manoj Kumar Yasangi, Recipient of Alberta Innovates Graduate Student Scholarship (GSS), $17,000

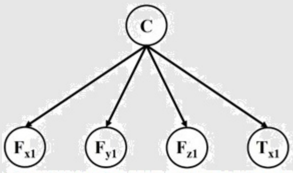
Master of Science (MSc) in Information Technology (In Progress)

Concordia University of Edmonton (CUE), Alberta, Canada

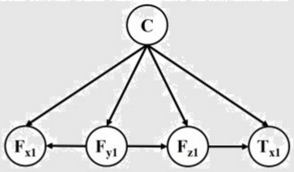
☐ Thesis title: *ECO-AI: Energy Consumption Optimization through Deep Learning Algorithms in Alberta*

❖ Present Position: *MSc Student, CUE*

BN

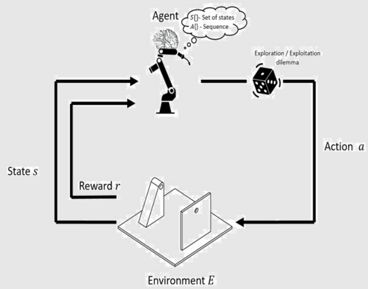


Naïve Bayes Structure

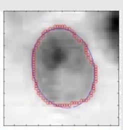


TAN Structure

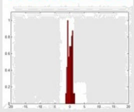
RL



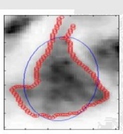
BRF



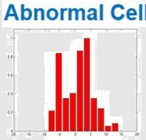
Normal Cell



Deviation from Ellipticity

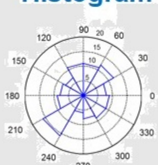


Abnormal Cell

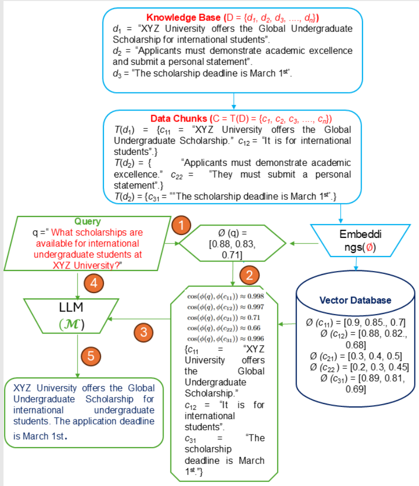


Deviation from Ellipticity

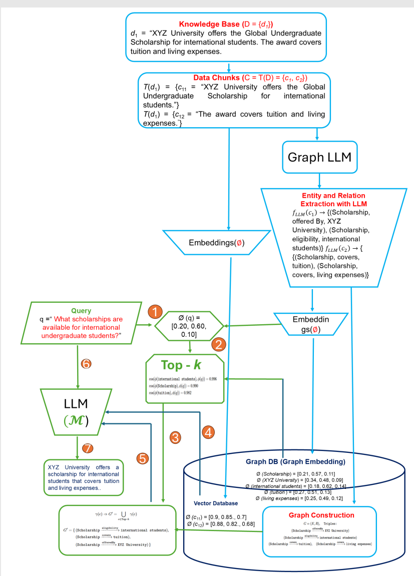
Angular Histogram



Retrieval-Augmented Generation (RAG)



Graph RAG



RAG vs Graph RAG

Feature	FAISS-RAG	GraphRAG
Handles complex relationships?	❌	✅ via structured graph
Accurate entity linking?	❌	✅
Explainability (trace reasoning)?	❌	✅ (graph paths are traceable)
Pure semantic search?	✅	✅ (on scoped results)
Use case: simple Q&A	✅	🟡 Overhead not needed
Use case: multi-entity filtered query	❌	✅

Fine-tune LLM and optimize RAG pipeline

Metric	LLM Base	RAG	GraphRAG
Relevance Score	0.6743	0.9919	—
BLEU	0.2875	0.2492	0.1762
ROUGE-1	0.5791	0.5407	0.4090
ROUGE-2	0.4239	0.4003	0.3307
ROUGE-L	0.4992	0.4560	0.3838
BERTScore Precision	0.3766	0.8735	0.8561
BERTScore Recall	0.6115	0.9271	0.9406
BERTScore F1	0.4914	0.8992	0.8961

Sep, 24 - Dec, 25 Kashvi Shah
Master of Science (MSc) in Information Technology (In Progress)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Wine Quality Prediction Through Multimodal Data Fusion*
❖ Present Position: *MSc Student, CUE*

Sep, 24 - Dec, 25 Varalakshmi Yasaswi Posina
Master of Science (MSc) in Information Technology (In Progress)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Graph Neural Network Models for Traffic Accident Prediction*
❖ Present Position: *MSc Student, CUE*

Sep, 24 - Dec, 25 Dattatreya Viswanadham
Master of Science (MSc) in Information Technology (In Progress)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Fusion-Based Multimodal Model for Wildfire Risk Prediction*
❖ Present Position: *MSc Student, CUE*

Sep, 24 - Dec, 25 Bhavithra Suruliyandi Selvakumar
Master of Science (MSc) in Information Technology (In Progress)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Space Debris Detection and Tracking Using Machine Learning and Computer Vision*
❖ Present Position: *MSc Student, CUE*

Sep, 24 - Dec, 24 Shwetaben Siddhpura, Recipient of Alberta Graduate Excellence Scholarship, $11,000
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Unlocking Data Insights with Retrieval-Augmented Generation (RAG)*
❖ Present Position: *Software Engineer*

May, 24 - Aug, 24 Sameera Nayyar Syeda
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Lung Cancer Detection Using Deep Learning Models*
❖ Present Position: *Lecturer, Engineering College, Hyderabad*

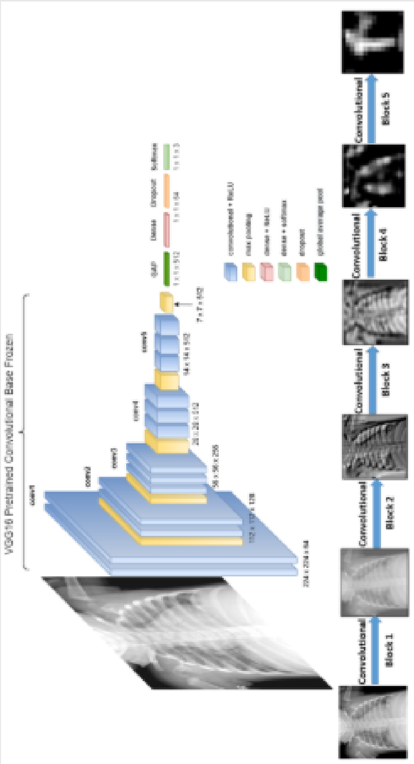
May, 24 - Aug, 24 Priyabahen Vaibhav Panchal
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Boosting LLM Performance with BLIP and OpenVINO for VQA and Image Captioning*
❖ Present Position: *Technical Consultant, Krish Technolabs*

May, 24 - Aug, 24 Aftab Ahmed
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Early Detection and Prediction of Forest Fires Using Deep Learning Techniques*
❖ Present Position: *Sales Representative, The Shoe Company*

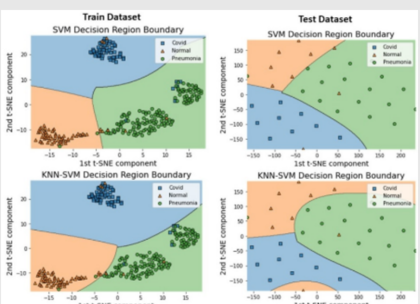
May, 24 - Aug, 24 Satyam Gaurangbhai Oza
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Multimodal Deep Learning for Skin Disease Classification*
❖ Present Position: *Software Engineer, Angel-Earth Corporation*

May, 24 - Aug, 24 Abhinav Aggarwal
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Security of Private Cloud Using Machine Learning and Cryptography*
❖ Present Position: *Technical Analyst, GoDaddy*

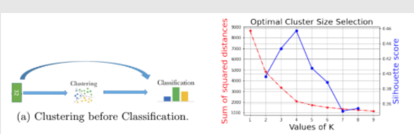
COVID 19 Induced Pneumonia Detection



Developed kNN-SVM Decision Boundary



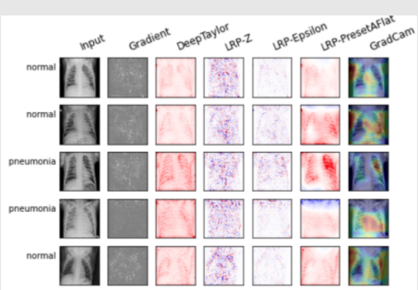
Clustering improves Classification



(a) Clustering before Classification.

(b) Optimal Values of K for K-Means Algorithm.

Representative chestXray correctly classified by kNN-SVM



Sep, 23 - Dec, 24 **Dharaben Wagh, Recipient of Alberta Innovates Graduate Student Scholarship (GSS), $17,000**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *AgriSustain: Advancing Soil Health and Sustainable Agriculture Using Machine Learning*
❖ Present Position: *Sessional Instructor, MacEwan University and CUE*

Sep, 23 - Dec, 24 **Pavan Sarvaiya, Recipient of Alberta Innovates Graduate Student Scholarship (GSS), $17,000**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *FrostGuard: Advancing Agriculture with AI and ML-based Frost Detection*
❖ Present Position: *Software Engineer, Angel-Earth Corporation*

Feb, 23 - Dec, 23 **Shahenoor Radhanpuri, Recipient of Alberta Innovates Graduate Student Scholarship (GSS), $17,000**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *AI Algorithms for Robotizing Manufacturing Task for Business Scale-Up and Productivity in Alberta*
❖ Present Position: *Software Engineer*

Sep, 22 - Dec, 23 **Tatva Joshi, Recipient of Alberta Graduate Excellence Scholarship, $11,000**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Deep Learning for Morphological Galaxy Classification and Exoplanet Detection: A Machine Learning Approach in Astrophysics*
❖ Present Position: *Data Analyst, Alberta Health Services*

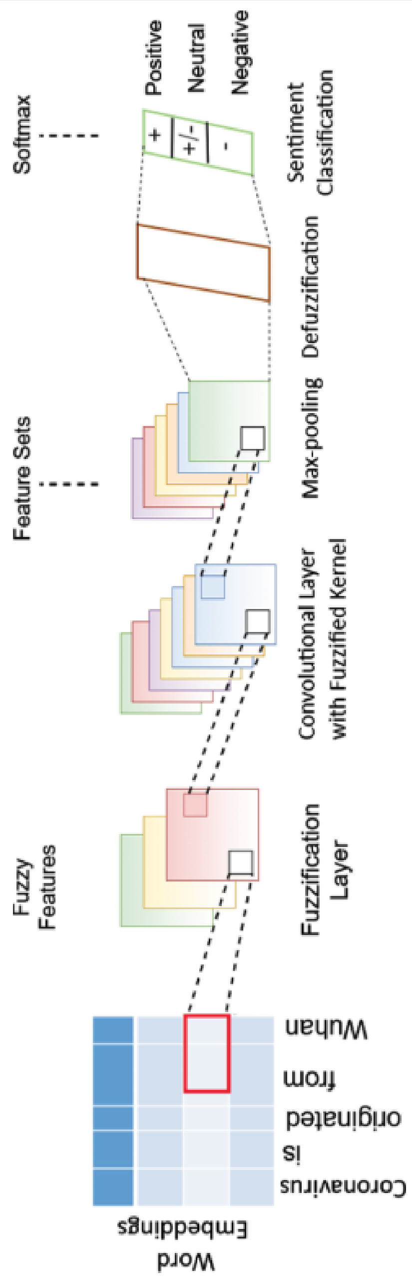
Sep, 22 - Dec, 23 **Ramya Nekkanti, Recipient of Alberta Innovates Graduate Student Scholarship (GSS), $17,000**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Machine Learning Based Models for Protein Structure Prediction in Drug Discovery*
❖ Present Position: *Data Engineer, Government of Alberta*

Sep, 22 - Dec, 23 **Md Khairul Anam**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Enhancing Movie Analysis: Multimodal Genre Classification and Personalized Recommendation in a Distributed Map Reduce Framework*
❖ Present Position: *Software Engineer*

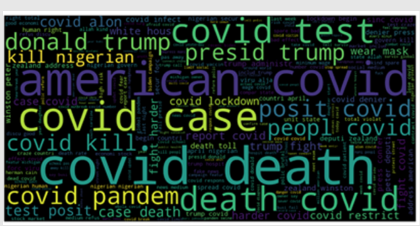
Sep, 22 - Dec, 23 **Shashank Reddy Panyala**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Multimodal Emotion Estimation Using Deep Learning Model*
❖ Present Position: *Software Engineer*

Sep, 22 - Dec, 23 **Archita Bhattacharya**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Addressing Ethical Considerations in Machine Learning Algorithms*
❖ Present Position: *Data Analyst, Toronto Dominion (TD) Bank*

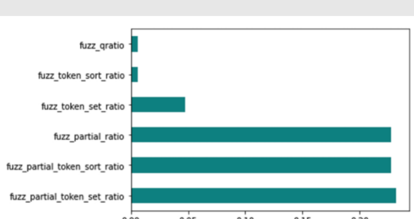
Deep Fuzzy Architecture



Covid WordCloud



Fuzzy Feature Selection



Sep, 22 - Dec, 23 **Md Ashraful Islam**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Unraveling the Radiogenomic Landscape of Glioblastoma Multiforme: A Multimodal Imaging and Transcriptomics Approach*
❖ Present Position: *IT Coordinator, Peerless College*

Jan, 22 - Dec, 22 **Wilson Kosasih**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Privacy Preserving Covid-19 Contact Tracing Using Machine Learning*
❖ Present Position: *Software Engineer*

Jan, 22 - Dec, 22 **Raj Patel**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Cell Image Detection and Recognition Using Deep Learning*
❖ Present Position: *Software Engineer*

Jan, 22 - Dec, 22 **Nabil Beshay**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Hate Speech Detection for Scarce Resource Languages*
❖ Present Position: *Software Engineer*

Sep, 21 - Dec, 22 **Raghavkumar Thakkar, Recipient of Governor General's Gold Medal**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Deep Active Contour Segmentation*
❖ Present Position: *Software Developer 2, Computronix*

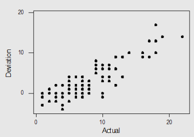
Sep, 21 - Dec, 22 **Kathan Trivedi, Recipient of Alberta Graduate Excellence Scholarship, $11,000**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Automated Weapon Detection and Recognition from Images Using Deep Learning*
❖ Present Position: *Software Engineer, Windowmaker Software Ltd*

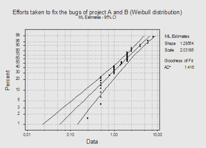
Sep, 21 - Dec, 22 **Dhruvilkumar Doshi**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Privacy Preserving Machine Learning*
❖ Present Position: *Software Developer, Computronix*

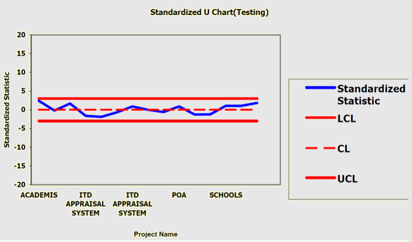
Sep, 21 - Dec, 22 **Nirmalkumar Laxmanbhai Patel**
Master of Science (MSc) in Information Technology (Completed)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Music Recognition by Fusing Video and Audio Signal*
❖ Present Position: *Senior Software Engineer, Persistent Systems*

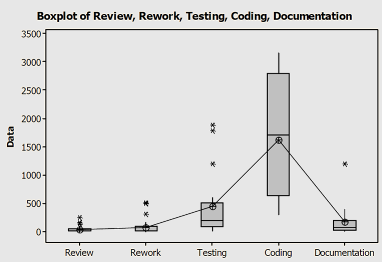
Sep, 21 -Dec, 22 **Ali Boukrich, Recipient of Canada Graduate Scholarships - Master's (CGS-M), $17,500**
Master of Science (MSc) in Information Technology (IT)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *A Unified View of Epidemiological Modelling and Artificial Intelligence-Based Approach to Understanding the COVID-19 Disease Spread for Supporting Public Policy Decisions*

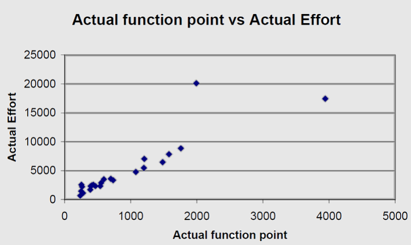
Software Effort Estimation



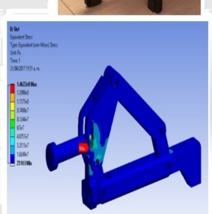


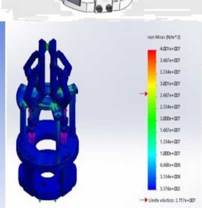


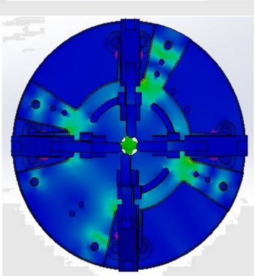




Gripper design using FEM







Sep, 20 - Dec, 21 **Rene Avalloni de Morais, Recipient of Alberta Graduate Excellence Scholarship, $11,000**
Master of Science (MSc) in Information Technology (IT)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Deep Learning Based Models for Software Effort Estimation Using Story Points in Agile Environments*
❖ Present Position: *Cloud Engineer, PointClickCare*

Sep, 20 - Dec, 21 **Qozeem Adeniyi Adeshina**
Master of Science (MSc) in Information Technology (IT)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Machine Learning Based Approach for Detecting Distributed Denial of Service Attack*
❖ Present Position: *Senior Technical Services Engineer, ZeroTek, Canada*

Sep, 20 - Dec, 21 **Adeola Ayandeyi, Recipient of Governor General's Gold Medal**
Master of Science (MSc) in Information Technology (IT)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Sentiment Analysis of Twitter Data to Analyze the Effect of Covid-19*
❖ Present Position: *Data Systems Analyst, Rogers Communications*

Sep, 20 - Dec, 21 **Victor Cheruiyot**
Master of Science (MSc) in Information Technology (IT)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Machine Learning Driven Software Test Case Selection*
❖ Present Position: *Senior Operations Administrator, FedEx Ground and Sessional Instructor, CUE*

Sep, 20 - Dec, 21 **Tanjimul Ahad Asif**
Master of Science (MSc) in Information Technology (IT)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title:*Retail System Automation with Machine Learning*
❖ Present Position: *QA Engineer II, IGT*

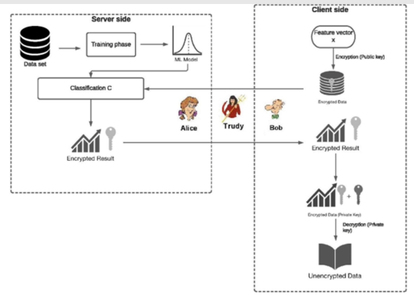
Sep, 20 - Dec, 21 **Lekhasree Narayanagari**
Master of Science (MSc) in Information Technology (IT)
Concordia University of Edmonton (CUE), Alberta, Canada
☐ Thesis title: *Developing Machine Learning Driven Internet of Things (IOT) based Smart Health Care Kit*
❖ Present Position: *Sessional Instructor, CUE*

Aug,16 - Jun, 21 **Adrían Alejandro Rodríguez Villarreal**
Master program in Computational Statistics
Centro de Investigación en Matemáticas (CIMAT), Monterrey, Mexico
☐ Thesis title: *Modelos de pronóstico de fallas basados en sensores para el monitoreo del estado de máquinaria utilizando Deep Learning (Sensor-based failure forecasting models for machinery health monitoring using Deep Learning)*
❖ Present Position: *Data Architect/Engineer, Neoris*

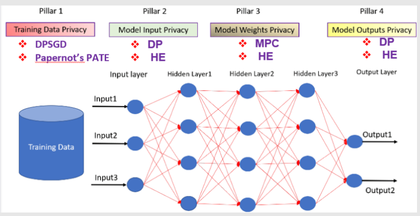
Aug, 16 - Feb, 19 **Arnulfo González Cantú**
Master program in Computational Statistics
Centro de Investigación en Matemáticas (CIMAT), Monterrey, Mexico
☐ Thesis title: *Decision Trees and their Application in Metabolic Syndrome*
❖ Present Position: *Endocrinologist, Vanguard Clinic, Monterrey, Mexico*

Aug, 15 -Apr, 19 **Jesús Elizalde Balboa**
Master program in Design and Development of Mechanical System
Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico
☐ Thesis title: *Design and analysis of reconfigurable clamping device for industrial robot*
❖ Present Position: *PD Engineer Sr., Centro de Ingeniería Navistar México*

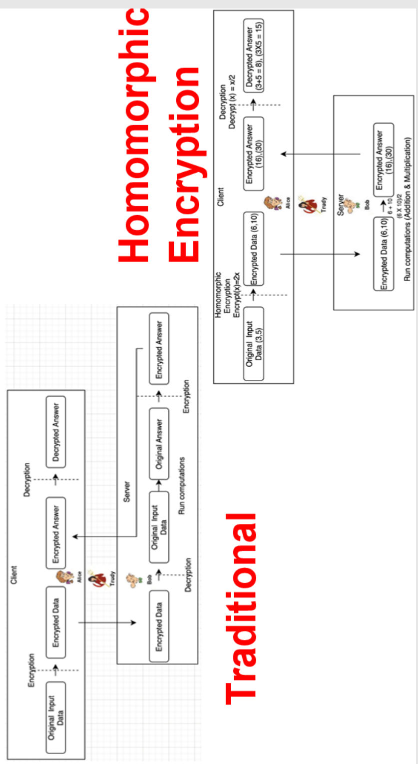
Privacy Preserving Machine Learning (PPML)



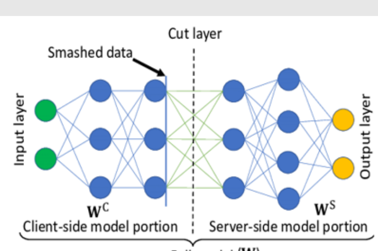
Four Pillars of PPML



Homomorphic Encryption



Split Network Learning



Undergraduate Students

Aug, 15 - Mar, 18 Jose Alejandro Alonso Tovar
Master program in Mechatronics
Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico
☐ Thesis title: *Design and implementation of a controller with force feedback for insertion task in a industrial robot*
❖ Present Position: *Leader of Autodesk Specialization Services, CiiSA - Consultoría Integral de Informática*

Sep, 24 - Dec, 24 Uloma Udeh — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐ Project title: *SymptoSync: An AI-Powered Platform for Managing PCOS and Women's Health*
❖ Present Position: *Founder, Symptosync*

Jan, 24 - Apr, 24 D S Mabrur Hyder — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐ Project title: *BotanicalD: Plant Identification and Optimal Growth Condition Analysis Tool*
❖ Present Position: *Technical Support Representative, Rogers Communications*

Jan, 24 - Apr, 24 Sanjot Singh — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐ Project title: *Fruit Detection and Nutritional Analysis Using Google Cloud Vision API and Machine Learning*
❖ Present Position: *Cashier, A&W Food Services of Canada Inc.*

Jan, 24 - Apr, 24 Vanessa Okyere — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐ Project title: *Developing an E-Commerce Platform for School Merchandise: A Data-Driven Approach Using HTML, CSS, and JavaScript*
❖ Present Position: *Cashier, Walmart*

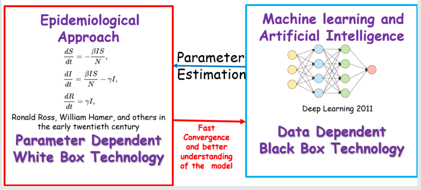
Jan, 24 - Apr, 24 Kosi Anadozie — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐ Project title: *Enhancing University Student Well-Being Through Digital Counseling Services and Mental Health Awareness Initiatives*
❖ Present Position: *Software Engineer*

May, 23 - Dec, 24 Gbenro Ketiku — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐ Project title: *Developing Intelligent Sports Prediction Analysis System (USRA, 2023) & Refining the Active Contour and Graph-Based Segmentation Through Deep Learning (USRA, 2024) &* Design and develop a mobile application for locating and reporting lost and found items within the university campus (Undergraduate Thesis)
❖ Present Position: *Quality Assurance Tester, Dough Dough Digital Inc.*

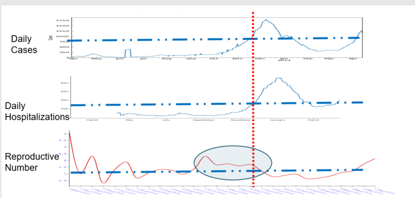
Jan, 23 - Apr, 23 Ashok Adhikari — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐ Project title: *Improved Multifactor Authentication with Machine Learning: A Novel Approach to Improve Safety*
❖ Present Position: *Software Engineer*

Jan, 23 - Apr, 23 Inderjot Kaur — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐ Project title: *Artificial Intelligence in Healthcare: Developing a Fitness App for Personalized Nutrition and Exercise Tracking*
❖ Present Position: *Assistant Manager, Kurves Beauty Bar*

Unifying Epidemiology and Deep Learning

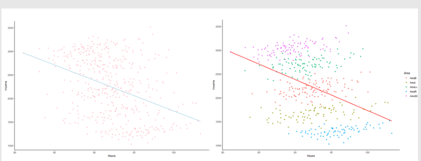


ML and Covid19 Early Detection

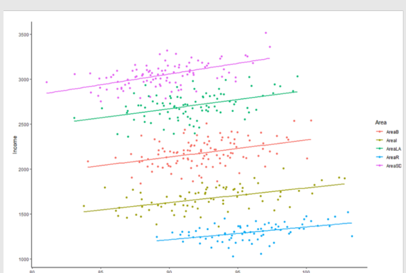


Marriage between traditional epidemiological Models and machine learning could help to develop novel parameters as early biomarker.

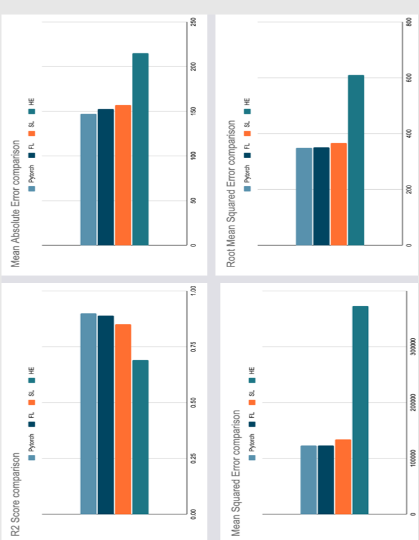
Association between different groups



Simpson Paradox and Linear Regression



PPML Model Evaluation



Jan, 23 - Apr, 23 **Tohidul Alam Anik** — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐Project title: *Mobile Apps for Fruit and Vegetable Detection to Improve Retail Store Billing System Automation Using Lightweight Deep Learning Models*
❖ Present Position: *System Consultant - Enoch Cree Nation, Long View Systems*

Jan, 23 - Apr, 23 **Ayo Okeowo** — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐Project title: *The Impact of Web Technologies on Daily Life: Developing a Weather Web-App Using JavaScript API*
❖ Present Position: *Data Analyst*

Sep, 21 - Jun, 22 **Gyan Ranjan** — Bachelor of Science (BSc) in Computer Science
Chandigarh University, India
☐Project title: *Sentiment Analysis from Text Data*
❖ Present Position: *SDE (AI/ML), AriveGuru Technology Solutions Pvt Ltd*

Sep, 21 - Apr, 22 **Rhythm Mahajan** — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐Project title: *Object Detection in Python Using OpenCV*
❖ Present Position: *Customer Experience Manager, The Home Depot Canada*

Sep, 21 - Apr, 22 **Manan Punjabi** — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐Project title: *Development of Key Logger Software and Machine Learning-Based Methods to Detect Unknown Key Loggers*
❖ Present Position: *IT Service Analyst, Shineatek Corp.*

Sep, 21 - Apr, 22 **Khoa Bui** — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐Project title: *A Reinforcement Learning-Based Algorithm for Developing Stacking Games*
❖ Present Position: *Technical Support Specialist, Canadian Securities Exchange (CSE)*

Jun, 21 - Jun, 22 **Deepak Yadav** — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐Project title: *Deep Learning for Audio Sentiment Analysis*
❖ Present Position: *Associate Lead Data Scientist, Bajaj Finserv*

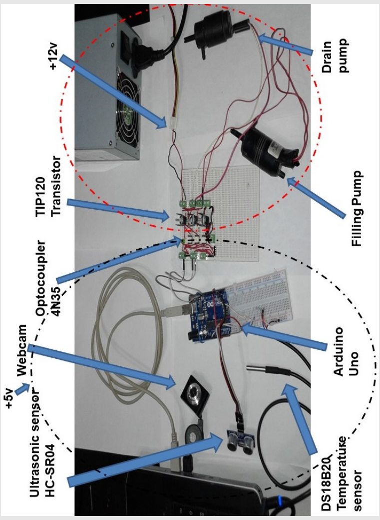
Jun, 21 - Jun, 22 **Aman Bahuguna** — Bachelor of Science (BSc) in Computer Science
Chandigarh University, India
☐Project title: *Deep Learning for COVID-Induced Pneumonia Detection from Chest X-ray*
❖ Present Position: *Student, Chandigarh University*

Sep, 20 - Aug, 21 **Shubham Malhotra** — Bachelor of Science (BSc) in Computer Science
Freelance, India
☐Project title: *COVID-19 Chatbot: A Natural Language Processing and Artificial Intelligence Powered Intelligent Virtual Assistant to Create Awareness During Pandemic*
❖ Present Position: *Production Associate, Canada Bread Company, Limited*

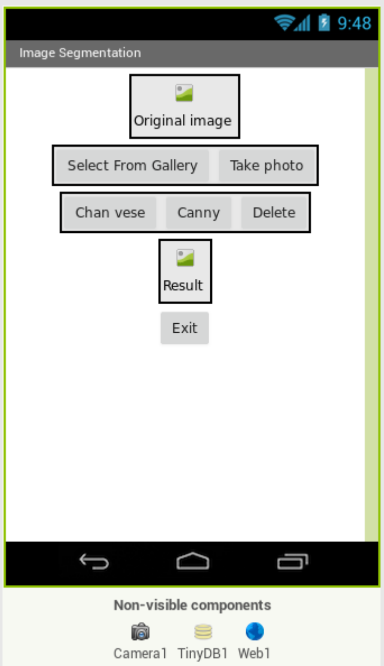
Sep, 20 - Aug, 21 **Jainth Chaudhary** — Bachelor of Science (BSc) in Computer Science
Pizza Hut, India
☐Project title: *Safe AI: Privacy Preserving Machine Learning Algorithms for Trust-Sensitive Environments*
❖ Present Position: *IT Technical Support, Election Alberta*

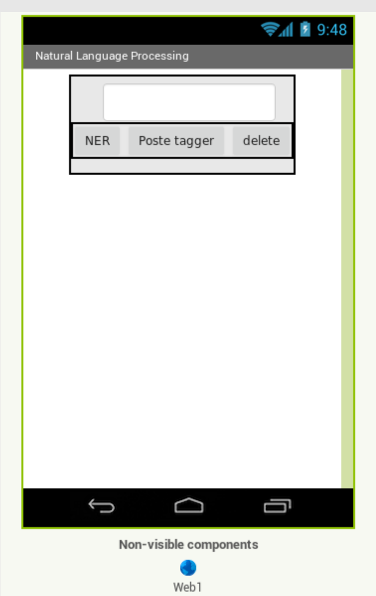
Sep, 20 - Aug, 21 **Shubhampreet Singh Manshahia** — Bachelor of Science (BSc) in Computer Science
Concordia University of Edmonton, Alberta, Canada
☐Project title: *Virtual Cook 2.0: A User Interactive Web-Based Smart Recipe Search Engine in Flask Framework With jQuery and Ajax*
❖ Present Position: *Software Developer, Elev, Edmonton*

Mechatronics



Mobile Apps





Aug, 14 - Jul, 18 Diana Laura Bañuelos Villegas
Bachelor of Industrial Processes Engineering
Universidad Autónoma de Sinaloa, Culiacán, Mexico
☐Project title: *Gas Metal Arc Welding (GMAW) sequence optimization for an agricultural component using artificial intelligence algorithms*
❖ Present Position: *Senior Information Product Manager, Salud Digna, Culiacán, Mexico*

Aug, 12 - Dec, 18 Luis Enrique Rodríguez González
Bachelor of Mechanical and Electrical Engineering
Universidad Autónoma de Nuevo León, Mexico and Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico
☐Project title: *Deep CNN for cell segmentation*
❖ Present Position: *ADAS Algorithms Engineer, Continental*

Aug, 11 - Jul, 16 Aranda Sinaí Miramontes
Mechatronics Engineering
Universidad tecnológica de Nayarit, Mexcio
☐Project title: *Welding sequence optimization using genetic algorithms*
❖ Present Position: *Project Engineer, Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico*

Aug, 12 - Apr, 17 Jonathan Ramiro Martinez Galin
Mechatronics Engineering
Instituto Tecnológico Superior de Poza Rica, Mexico and Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico
☐Project title: *Machine learning techniques for the "Peg-In-Hole" task*
❖ Present Position: *Senior Continuous Engineer Support, Continental, Queretaro, Mexico*

Aug, 13 - May, 17 Harsh Verma
Bachelor of Technology in Mechanical Engineering
National Institute of Technology, Patna, India
☐Project title: *Job scheduling problem for "N" jobs and "N" machines Using First Come, First Serve (FCFS) algorithm with backtracking*
❖ Present Position: *Senior Analyst, Capgemini, Bangalore, India*

Aug, 13 - May, 17 Gauravdeep
Bachelor of Technology in Mechanical Engineering
National Institute of Technology, Patna, India
☐Project title: *Job scheduling problem for "N" jobs and "N" machines Using First Come, First Serve (FCFS) algorithm with backtracking*
❖ Present Position: *Engineer, Larsen & Toubro (L & T)*

Aug, 12 - Apr, 17 David Adrían Beltran Borquez
Mechatronics Engineering
Instituto Tecnológico Nacional de Mexico, Los Mochis, Sinaloa, Mexico
☐Project title: *Application of artificial intelligence algorithms for welding sequence optimization*
❖ Present Position: *Engineer, Culiacán, Mexico*

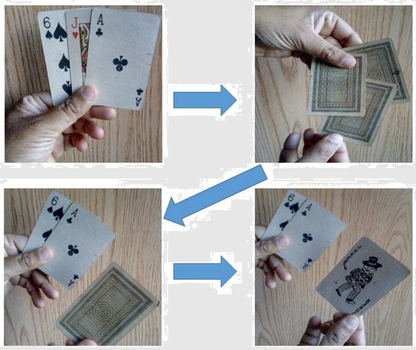
Summer Students

Aug, 14 - Jul, 18 Debopriyo Bhattacharya
Bachelor of Computer Science and Engineering
Heritage Institute of Technology, Kolkata, India
☐Project title: *Parallelization of Active Contour based segmentation algorithms*

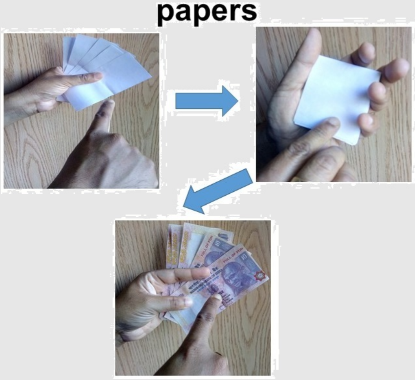
Aug, 13 -July, 18 Ernesto Salazar Joya
Mechatronics Engineering
Instituto Tecnológico de Culiacán, Sinaloa, Mexico
☐Project title: *A Reinforcement Learning (RL) based welding sequence optimization method for a mechanical component*

Magic Demonstrated

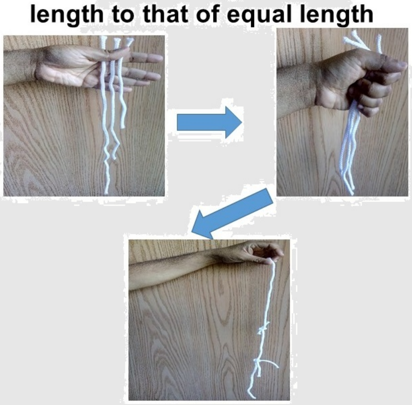
Vanish the red card



Making money from white papers



Making three strings of unequal length to that of equal length



Vanish a set of coins from the coin box

Aug, 11 - Dec, 16 Andrés Sauceda Cienfuegos
Bachelor of Mechanical and Electrical Engineering
Universidad Autónoma de Nuevo León, Mexico and Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico
☐Project title: *Computer vision based robot automation for peg in hole insertion task*

Aug, 11 - Dec, 16 Dario Alberto Cruz Santiago
Bachelor of Mechatronics Engineering
Instituto Tecnológico Superior de Poza Rica, Veracruz, Mexico and Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico
☐Project title: *Computer vision based algorithm for automatic detection of the contact points of the gripper for lifting objects of irregular shapes through Genetic Algorithm*

Aug, 11 - Dec, 16 Jesus Eloradana Ochoa Corona
Bachelor of Industrial Processes
Universidad Autónoma de Sinaloa, Culiacán, Mexico and Centro de Ingeniería y Desarrollo Industrial (CIDESI), Monterrey, Mexico
☐Project title: *Detection and removal of chattering during drilling operation*

Aug, 11 - Dec, 16 Saúl Herrera
Bachelor of Physics
Universidad Autónoma de Nuevo León, Monterrey, Mexico
☐Project title: *Developing Support Vector Machines for big data*

Aug, 11 - Dec, 16 Daniel Cavazos
Bachelor of Physics
Universidad Autónoma de Nuevo León, Monterrey, Mexico
☐Project title: *Developing Support Vector Machines for big data*

Aug, 13 - July, 18 Ethan Nava López
Mechatronics Engineering
Instituto Tecnológico de Culiacán, Sinaloa, Mexico
☐Project title: *An Arduino based controller for regulating the level and temperature of the tank water using the combination of mulitple sensors (ultrasonic, camera, and temperature) with Kalman filter*

Aug, 13 - July, 18 Jesús Ernesto Osuna Jiménez
Mechatronics Engineering
Instituto Tecnológico de Culiacán, Sinaloa, Mexico
☐Project title: *A Raspberry based line tracker using robot in the presence of obstacles using the combination of mulitple sensors (infrared, ultrasonic, and camera) with Kalman filter*

Aug, 13 - July, 18 Dorian Alexis Velazquez Sotomayor
Mechatronics Engineering
Instituto Tecnológico de Colima, Mexico
☐Project title: *Developing algorithms to segment and separate overlapping cells through Chan-Vese algorithm*

External Examiner & Article Reviewer

Served as an external PhD examiner and reviewer for 70+ articles across top journals and conferences, including:
❖ Emergeing Applications of Artificial Intelligence (EAAI), 2021
❖ International Conference on Machine Intelligence and Signal Processing (MISP), 2021
❖ IEEE Transactions Image Processing, Pattern Recognition
❖ IEEE Transactions of Man, Machine and Cybernatics, Part-C
❖ IEEE Conference of Computer Vision and Pattern Recognition (CVPR), 2014